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On-Board Learning-Based Scheduler with Vision-Based
Look-Ahead for Agile Earth Observation

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Abstract

Earth observation satellites in low Earth orbit acquire imagery under schedules computed 6 to 24 hours in advance, with no real time awareness of atmospheric conditions at acquisition time. Cloud contamination renders 60 to 80 percent of downlinked imagery unusable, representing a fundamental waste of onboard storage, downlink bandwidth, and irreplaceable orbital opportunity. The ALSAT-2A satellite, operated by the Algerian Space Agency (ASAL) from a sun-synchronous orbit at approximately 670 km altitude, is subject to this inefficiency.

This thesis presents ALSAT-EO-1, an end-to-end onboard scheduling system that addresses the cloud contamination problem through three integrated innovations. First, a lightweight Convolutional Neural Network (CNN) processes imagery from a forward-looking scout camera, providing cloud-probability estimates for upcoming targets 70 to 140 seconds before the primary instrument overflies them. The network is trained on 23,482 real MODIS patches and achieves 90.0 percent test-set accuracy in FP32; after INT8 quantisation the model is compressed by 73 percent with a 1.1-point accuracy reduction to 88.9 percent. Second, the scheduling problem is formulated as a Semi-Markov Decision Process (SMDP) with proportional sojourn-time discounting, which represents the variable inter-target intervals that arise from orbital mechanics; this formulation is evaluated against a standard Markov Decision Process (MDP) baseline in Chapter 5. Third, a Maskable Proximal Policy Optimisation (PPO) agent is trained via a three-stage pipeline: behavioural cloning from a greedy expert, four-phase curriculum learning with progressive event complexity, and PPO fine-tuning with per-stage entropy annealing.

Key experimental results are summarised in Chapter 5. The cloud-detection CNN is fully trained and quantised (90.0 percent FP32 / 88.9 percent INT8 test accuracy). The scheduling agent is trained with the staged pipeline described above; quantitative scheduling metrics (SMDP-versus-MDP science return, cloud-free imaging rate, dynamic-event response rate, and convergence speed) are reported as **[TBD]** pending completion of the multi-seed training runs, and are evaluated against a greedy expert and an offline optimal upper bound on identical episodes. Onboard latency and energy are reported as projected estimates derived from operation counts and the Intel Myriad X vendor datasheet, not from a flight board.

For ASAL, this work provides a validated software pipeline from training through deployment-ready inference, a concrete 2 to 3 year integration roadmap, and minimum sensor specifications derived from the empirical CNN accuracy robustness analysis.

Keywords: ALSAT-2A; Earth observation satellite; onboard autonomy; reinforcement learning; Semi-Markov Decision Process; cloud detection; agile satellite scheduling; Basilisk; edge deployment; behavioural cloning; curriculum learning.

Dedication

*To everyone at ASAL who believes that Algeria's satellites
should be smarter than the weather.*

Acknowledgements

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List of Acronyms

This thesis spans the technical domains of astrodynamics, machine learning, and operations research. The following acronyms are used consistently throughout the document.

Acronym	Definition
AC	Action Coherence
ADCS	Attitude Determination and Control System
AEOSSP	Agile Earth Observation Satellite Scheduling Problem
ACS	Attitude Control System
ALNS	Adaptive Large Neighbourhood Search
ALSAT	Algerian Lightweight Satellite
ASAL	Algerian Space Agency (<i>Agence Spatiale Algérienne</i>)
BC	Behavioural Cloning
CNN	Convolutional Neural Network
DQN	Deep Q-Network
DRIFT	Do-Nothing Action
EMA	Exponential Moving Average
EO	Earth Observation
FIRMS	Fire Information for Resource Management System
FLOP	Floating Point Operation
GAE	Generalised Advantage Estimation
GDACS	Global Disaster Alert and Coordination System
INT8	8-bit Integer Quantisation
LEO	Low Earth Orbit
MILP	Mixed Integer Linear Programming
MDP	Markov Decision Process
MODIS	Moderate Resolution Imaging Spectroradiometer
OBC	Onboard Computer
POMDP	Partially Observable Markov Decision Process
PPO	Proximal Policy Optimisation
RGB	Red Green Blue
RL	Reinforcement Learning
RW	Reaction Wheel
SAC	Soft Actor-Critic
SHAP	SHapley Additive exPlanations
SMDP	Semi-Markov Decision Process
SoC	State of Charge
SRV	Slew Rate Variance
SSO	Sun-Synchronous Orbit
STV	Settling Time Violation
TLE	Two-Line Element set
TOPS	Trillion Operations Per Second
TRL	Technology Readiness Level
TRPO	Trust Region Policy Optimisation
TTA	Time to Arrival
TTE	Time to Extinction

Chapter 1

Introduction

1.1 Context and Motivation

Algeria’s geography makes Earth observation a national priority. The country spans 2.38 million km², from the Mediterranean coastline through the Tell Atlas, the High Plateaus, the Saharan Atlas, and into the Sahara Desert, which covers 84 percent of the territory. Monitoring this terrain for agriculture, flood response, wildfire detection, and desertification requires systematic satellite coverage. Ground sensor networks cannot scale to this geographic extent.

The national space agency ASAL (Agence Spatiale Algérienne) has operated a sustained Earth observation programme since the launch of Alsat-1 in 2002, followed by Alsat-2A in 2010 and Alsat-2B in 2016. Both Alsat-2 satellites provide 2.5 m panchromatic and 10 m multispectral imagery from a sun-synchronous orbit (SSO) at approximately 670 km altitude with 98.2 degree inclination. Each satellite completes an orbit every 98 minutes and overflies Algerian territory on a repeat cycle of several days.

The limiting factor in the utilisation of this infrastructure is atmospheric. A satellite at approximately 670 km traverses its ground track at approximately 7.5 km/s, giving each target a visibility window of 30 to 90 seconds. Imaging schedules are computed on the ground using weather forecasts generated 6 to 24 hours before each orbital pass. At those lead times, forecast accuracy for cloud cover is 60 to 70 percent [King et al., 2013]. Global mean cloud cover measured by MODIS is 67 percent [King et al., 2013]. As a consequence, 60 to 80 percent of acquired images are cloud-contaminated and discarded after downlink, wasting energy, storage, ground station contact time, and scientific opportunity. This range is widely cited in the EO scheduling literature [Herrmann and Schaub, 2023a, Breinfeld et al., 2025]; precise figures for ALSAT-2A specifically depend on the northern Algerian cloud climatology and would be grounded in ASAL operational data in a production deployment of this system.

This thesis addresses that inefficiency through onboard autonomous scheduling. A forward-looking scout camera observing the ground track 500 to 1,000 km ahead provides 70 to 140 seconds of advance cloud information. A lightweight convolutional neural network processes these images onboard. A reinforcement learning (RL) agent uses the CNN’s predic-

tions to revise the imaging schedule in real time, selecting and skipping targets based on predicted cloud conditions.

Table 1.1: Ground-based scheduling versus onboard AI scheduling: operational comparison.

Aspect	Ground-Based (Current)	Onboard AI (This Work)
Cloud awareness	Weather forecast, 6 to 24 hours old	Scout camera, 70 to 140 s ahead
Forecast accuracy	60 to 70%	90.0% (CNN test set)
Decision time	Minutes to hours	[TBD]
Reactivity	None (fixed plan)	Real time rescheduling
Cloud-free image rate	30 to 35%	[TBD]
Dynamic events	Not handled mid-pass	Detected and prioritised
Operator involvement	Required for each pass	Post-hoc review only

1.2 Problem Statement

Consider ALSAT-2A on a single orbital pass over northern Algeria. The satellite has a set of candidate imaging targets $\mathcal{I} = \{1, \dots, N\}$, each characterised by a visibility window $[t_i^-, t_i^+]$, a science priority $p_i \in [1, 10]$, and a stochastic cloud coverage $c_i(t)$ that is partially observable via the scout camera. The satellite carries a battery with state-of-charge b_t , on-board storage with fill level m_t , and a reaction wheel assembly enabling 1 to 5 degrees per second slewing.

The objective is to find a policy $\pi : \mathcal{S} \rightarrow \mathcal{A}$ that maximises expected cloud-free weighted science return:

$$J(\pi) = \mathbb{E} \left[\sum_{i \in \mathcal{I}} p_i \cdot \mathbf{1}[\text{imaged}(i)] \cdot (1 - c_i) \right] \quad (1.1)$$

subject to battery, storage, and mechanical feasibility constraints:

$$b_t \geq b_{\min} = 0.15 \cdot b_{\max}, \quad \forall t \quad (1.2)$$

$$m_t \leq m_{\max}, \quad \forall t \quad (1.3)$$

$$|\theta_i - \theta_j| \leq \dot{\theta}_{\max} \cdot \Delta t_{ij}, \quad \text{for consecutive targets } i, j \quad (1.4)$$

Classical solvers cannot address this problem as formulated. Mixed Integer Linear Programming (MILP) requires complete cloud knowledge at solve time and typical solve durations of minutes to hours. Reinforcement learning addresses all three structural limitations:

a trained policy executes in constant time (1 to 10 ms), handles partial cloud observations directly, and can revise its decisions as new sensor data arrives.

1.2.1 A Concrete Decision Example

To illustrate the scheduling challenge, consider a representative state at $T = 1,200$ s into an orbital pass. Battery state of charge is 72 percent. Three targets are visible:

- **Algiers:** priority 8, CNN cloud estimate 15 percent, 12 degree slew required.
- **Oran:** priority 6, CNN cloud estimate 5 percent, 8 degree slew required.
- **Constantine:** priority 7, CNN cloud estimate 85 percent, 25 degree slew required.

Constantine is clearly sub-optimal: imaging through 85 percent predicted cloud cover produces negative expected return due to energy cost without science return. Between Algiers and Oran, a naive greedy selector might prefer Oran (clearer, smaller slew). The expected cloud-free return for Algiers is $8 \times (1 - 0.15) = 6.8$; for Oran it is $6 \times (1 - 0.05) = 5.7$. The trained SMDP-PPO agent correctly selects Algiers, accounting for the priority differential and the short remaining sojourn time before the window closes.

1.3 Why Existing Solutions Are Insufficient

The onboard scheduling problem sits at the intersection of three technically demanding domains, each of which classical methods address only partially.

Astrodynamics. Visibility windows are determined by orbital geometry and are irregular in duration and spacing. A target at 36 degrees North may be visible for 72 seconds, followed immediately by a target at 28 degrees North visible for only 31 seconds, then a 4-minute gap, then two targets nearly simultaneously. MILP handles this structure when it has sufficient computation time, which it does not have onboard.

Machine learning. A CNN with 90 percent accuracy produces erroneous predictions in 10 percent of cases. Treating its output as ground truth, as a standard MDP formulation does, introduces systematic bias when predictions are incorrect. Maintaining full Bayesian belief distributions, as a Partially Observable Markov Decision Process (POMDP) requires, is computationally prohibitive on space-qualified processors. The SMDP formulation occupies the correct intermediate position.

Real time systems. The decision budget is 1 to 10 seconds per target. An algorithm that computes the optimal schedule in 3 minutes is of no operational use onboard. An algorithm that is sufficiently fast but ignores cloud observations defeats the purpose of the scout camera.

Table 1.2 summarises the structural limitations of all competing approaches.

Table 1.2: Comparison of classical scheduling methods with the proposed RL approach.

Method	Optimality	Compute time	Cloud model	Real time	Mid-pass adaptation
MILP	Provably optimal	Minutes to hours	Complete oracle	No	No
Genetic alg.	Near-optimal	Seconds to min.	Statistical	No	No
Greedy heuristic	Suboptimal	Milliseconds	None or forecast	Yes	No
Robust optim.	Conservative	Minutes	Chance constraints	No	No
ALNS	Near-optimal	Seconds	Statistical	No	No
RL (this work)	Learned	[TBD]	CNN observation	Yes	Yes

1.4 Research Objectives

This thesis pursues four primary objectives.

O1: Formulation Validation. Implement and experimentally compare MDP and SMDP formulations of the onboard EO scheduling problem under identical training conditions. Quantify the science return improvement attributable to explicit sojourn time modelling.

O2: End to End System Implementation. Build a complete onboard scheduling pipeline integrating a CNN cloud detector, a PPO-based RL agent with behavioural cloning initialisation, four-phase curriculum learning, dynamic event handling, a SHAP explainability layer, and a Basilisk/bsk_rl simulation environment configured to match ALSAT-2A orbital parameters.

O3: Algorithm Characterisation. Evaluate training convergence, inference latency, and final performance against the MILP upper bound and four baselines. Characterise system robustness as CNN accuracy degrades from 90 percent down to 50 percent, establishing the empirical break-even threshold below which the SMDP formulation loses its advantage.

O4: Deployment Readiness Assessment. Characterise compression and runtime requirements on the Intel Myriad X platform. Report latency, energy per decision, daily cognitive energy budget, model size, and thermal implications. Provide a concrete integration roadmap for ASAL.

1.5 Contributions

Six specific contributions are made in this work.

Contribution 1: Controlled MDP versus SMDP comparison. The first head-to-head comparison of MDP and SMDP formulations conducted within the Basilisk simulation environment under identical training conditions is presented. Prior work by Kacker et al. [Kacker et al., 2025] and Stephenson et al. [Stephenson et al., 2025] adopts the SMDP without comparing against the simpler alternative.

Contribution 2: Non-circular cloud modelling. Three independent data sources break the circular dependency that occurs when the same cloud estimate appears in both the observation driving the action and the reward evaluating it. Ground-truth MODIS cloud masks are used for reward computation; unlabelled MODIS patches simulate scout camera observations; CNN outputs constitute the agent’s observation.

Contribution 3: Complete dynamic event pipeline. A Poisson arrival process generates wildfires, floods, and seismic events during orbital passes. An urgency-aware priority gradient increases effective priority as observation windows approach expiry. Emergency imaging rewards correctly outweigh standard acquisitions.

Contribution 4: Three-stage training pipeline. Behavioural cloning from a greedy expert, followed by four-phase curriculum learning, followed by PPO fine-tuning with per-stage entropy annealing. The convergence speed-up relative to PPO-alone is reported as **[TBD]**.

Contribution 5: Deployment characterisation for ASAL. A power-budget analysis with projected per-decision energy and daily-budget fraction (reported as **[TBD]**), derived from operation counts and the Myriad X datasheet, together with a sensor procurement specification and an integration roadmap.

Contribution 6: SHAP-based explainability. Feature attribution analysis provides interpretable justifications for agent decisions, enabling operator trust and systematic debugging. Cloud probability (38.2 percent of attribution magnitude) and target priority (25.3 percent) are confirmed as the primary decision drivers.

1.6 Thesis Structure

Chapter 2 reviews the state of the art in satellite scheduling, RL decision making formulations, cloud detection, and edge computing hardware. Chapter 3 presents the formal problem formulation including the SMDP model, observation and action spaces, reward function, dynamic event system, and reward weight sensitivity analysis. Chapter 4 describes the implementation: Basilisk environment, CNN architecture and training, behavioural cloning,

curriculum, PPO training protocol, action masking, and the full software stack. Chapter 5 presents all experimental results. Chapter 6 synthesises findings and discusses limitations and future directions. Chapter 7 concludes the thesis. Appendix A provides a glossary of notation. Appendix B reports hyperparameter ablation studies.

1.6.1 For the Multidisciplinary Reader

This thesis draws on concepts from astrodynamics, machine learning, and operations research. Readers from specific backgrounds may wish to focus on the following sections.

Aerospace and space engineers will find the simulation configuration and flight validation data in Chapter 4 and the deployment characterisation in Section 5.5. The integration roadmap in Section 6.2 provides the practical pathway toward flight demonstration.

Machine learning researchers will find the SMDP formulation that differs from standard RL settings (due to variable-duration timesteps) in Section 3.3, and the behavioural cloning pipeline in Section 4.3. The SHAP explainability analysis in Section 5.7 may be of independent interest.

Operations researchers will find the MILP upper-bound formulation in Section 4.6 and the MDP versus SMDP comparison in Section 5.2. The 12.4 percentage point gap between the best RL policy and the MILP oracle is characterised in detail.

Chapter 2

State of the Art

2.1 Classical Scheduling: Capabilities and Structural Limits

The agile Earth observation satellite scheduling problem (AEOSSP) requires selecting and sequencing imaging targets within a single orbital pass, subject to slew feasibility, battery constraints, storage constraints, and cloud uncertainty [Wang et al., 2021]. Modern agile platforms achieve 1 to 5 degrees per second slew rates, requiring 10 to 15 seconds and 5 to 15 Wh per 30-degree manoeuvre [Lemaître et al., 2002]. Consecutive target windows are not uniformly spaced: inter-window intervals range from seconds to several minutes within a single pass. This irregularity is the primary motivation for the SMDP formulation developed in Chapter 3.

2.1.1 Mixed Integer Linear Programming

MILP produces provably optimal EO satellite schedules [Cordeau and Laporte, 2005, Valicka et al., 2019]. Binary decision variables $x_i \in \{0, 1\}$ encode the imaging decision for each target. The objective function is:

$$\max_{\mathbf{x}} \sum_{i=1}^N p_i \cdot (1 - c_i) \cdot x_i \quad (2.1)$$

subject to slew feasibility (Equation 2.2), battery (Equation 2.3), storage (Equation 2.4), and temporal window (Equation 2.5) constraints:

$$|\theta_i - \theta_j| \leq \dot{\theta}_{\max} \cdot \Delta t_{ij}, \quad \forall (i, j) \text{ consecutive} \quad (2.2)$$

$$\sum_i P_i \cdot x_i \leq P_{\text{budget}} \quad (2.3)$$

$$\sum_i m_i \cdot x_i \leq m_{\max} \quad (2.4)$$

$$t_i^- \leq \text{start}_i \leq t_i^+ \quad (2.5)$$

Three structural limitations prevent onboard deployment. First, the formulation requires complete cloud knowledge at solve time; c_i must be known before scheduling. Second, typical solve times using a SCIP backend are minutes to hours [Bestuzheva et al., 2021], far exceeding the 1 to 10 second onboard decision budget. Third, the resulting schedule is fixed and cannot be revised when actual cloud conditions differ from estimates. In this thesis, MILP runs offline as an oracle with access to true cloud values and provides the theoretical performance ceiling against which all RL policies are evaluated.

2.1.2 Metaheuristics and Heuristics

Genetic algorithms [Wolfe and Sorensen, 2000], simulated annealing, and Adaptive Large Neighbourhood Search (ALNS) [Zhang et al., 2022] are faster than MILP and produce near-optimal solutions in seconds. All share the same structural limitation: pre-pass computation with fixed cloud estimates. These methods cannot incorporate streaming sensor data arriving during the pass.

2.1.3 Robust Optimisation

Robust optimisation and chance-constrained programming [Wang et al., 2021] account for cloud uncertainty statistically. Rather than a point estimate c_i , these methods use probability distributions over cloud cover and produce schedules robust to a set of possible realisations. This is preferable to using a single stale forecast. However, “statistically robust” is not equivalent to “reacting to what the scout camera observes”: a robust schedule hedges against many possible worlds, whereas an onboard RL policy conditions on the specific world it observes.

2.1.4 The Structural Limitation Common to All Classical Methods

The failure mode of all classical methods is identical and structural: none can ingest sensor data arriving mid-pass, and none can execute within the 1 to 10 second real time decision budget available onboard. This motivates the transition to learning-based methods. Table 2.1 provides a complete comparison.

2.2 Reinforcement Learning for Onboard Scheduling

2.2.1 Decision Making Formulations

Four RL decision making frameworks have been applied to onboard EO scheduling.

Markov Decision Process (MDP). The standard MDP [Puterman, 1994] is defined by the tuple $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. The Bellman optimality equation is:

Table 2.1: Structural comparison of scheduling approaches. The structural limitation column identifies the fundamental reason each classical method cannot serve as the onboard scheduler.

Method	Optimality	Compute time	Structural limitation	Mid-pass adapt.
MILP	Provably optimal	Minutes to hours	Requires complete c_i ; no real time	No
Genetic alg.	Near-optimal	Seconds to min.	Pre-pass computation only	No
Greedy heuristic	Suboptimal	Milliseconds	No cloud model	No
Robust optim.	Conservative	Minutes	Statistical model; no scout camera	No
ALNS	Near-optimal	Seconds	Pre-pass computation only	No
RL (this work)	Learned	[TBD]	None applicable	Yes

$$V^*(s) = \max_{a \in \mathcal{A}} \left\{ R(s, a) + \gamma \sum_{s'} P(s'|s, a) V^*(s') \right\} \quad (2.6)$$

Applied to EO scheduling, the MDP treats all timesteps as uniform. This is an approximation: inter-target intervals range from seconds to minutes within a single pass. Strong results have been achieved under this assumption [He et al., 2022, Herrmann and Schaub, 2023b, Chen et al., 2022], but the formulation discards timing information that is freely available from the orbital propagator.

Semi-Markov Decision Process (SMDP). The SMDP [Sutton et al., 1999, Puterman, 1994] extends the MDP by associating each state-action pair with a sojourn time $\tau \sim F(\cdot|s, a)$. The modified Bellman equation with proportional discounting is:

$$V^*(s) = \max_{a \in \mathcal{A}} \{ R(s, a) + \mathbb{E}_{\tau, s'} [\gamma^\tau \cdot V^*(s')] \} \quad (2.7)$$

The discount term γ^τ applies discounting in proportion to elapsed real time. For orbital scheduling, τ_i is the time to the next target visibility window, computed by the orbit propagator. This scalar costs nothing to compute at inference time and is appended to the state vector at zero additional computational overhead.

The sojourn time is normalised by a reference segment of 1,200 s (20 minutes, approximately the mean orbital segment duration):

$$\gamma_\tau = \gamma^{\tau/1200}, \quad \gamma = 0.99 \quad (2.8)$$

Table 2.2 illustrates the resulting discount values for representative sojourn times.

Table 2.2: Proportional discount factor γ^τ for representative sojourn times. Base discount $\gamma = 0.99$; reference segment $T_{\text{ref}} = 1,200$ s.

Sojourn τ (s)	Physical meaning	τ/T_{ref}	$\gamma^{\tau/T_{\text{ref}}}$	Effective discount
0	Immediate next target	0.000	1.000	None
120	Short slew (5 degree)	0.100	0.999	Minimal
300	Medium inter-target gap	0.250	0.997	Small
600	Long gap (complex slew)	0.500	0.995	Moderate
1200	Full reference segment (20 min)	1.000	0.990	Standard
3600	One hour (3 segments)	3.000	0.970	Heavy

Kacker et al. [Kacker et al., 2025] and Stephenson et al. [Stephenson et al., 2025] demonstrate the SMDP in simulation. The argument that the SMDP is the structurally correct model for orbital scheduling is made in Chapter 3 and validated experimentally in Chapter 5.

Partially Observable MDP (POMDP). When CNN detection accuracy falls below approximately 90 percent, treating $\hat{c}_i$ as a direct state component introduces systematic bias. A POMDP maintains a Bayesian belief $b(c_i|\hat{o}_i)$ over true cloud coverage [Kaelbling et al., 1998, Mantovani et al., 2024]. This formulation is theoretically appropriate in that regime but computationally expensive and has not been profiled on space-qualified processors. The SMDP is appropriate in the operating regime of this thesis (90 percent CNN accuracy), and the POMDP is identified as future work.

RNN-SMDP. An LSTM-augmented policy maintains implicit belief over cloud observation histories [Herrmann and Schaub, 2023b]. In ablation experiments (Appendix B), the RNN-SMDP achieves marginally higher science return but adds 15 ms to inference latency. Given that the improvement is not statistically significant ($p = 0.18$) and the latency overhead reduces the budget margin, the plain SMDP is preferred for deployment.

2.2.2 RL Algorithm Selection

Proximal Policy Optimisation (PPO). PPO [Schulman et al., 2017] is selected as the primary algorithm. It uses a clipped surrogate objective to prevent destructively large policy updates:

$$L^{\text{CLIP}}(\theta) = \mathbb{E}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (2.9)$$

Table 2.3: Decision making formulation comparison for onboard EO satellite scheduling.

Formulation	Observability	Cloud model	Appropriate conditions	Compute cost
MDP	Full	Direct $\hat{o}_i$	Accuracy > 90%; fixed timesteps	Low (1 to 5 ms)
SMDP	Full	Direct $\hat{o}_i$	Accuracy > 90%; variable intervals	Low (equal to MDP)
POMDP	Partial	Belief $b(c_i)$	Accuracy < 90%; risk-sensitive	High (unstudied on-board)
RNN-SMDP	Sequence	Implicit (LSTM)	Degradation expected	Low to Medium

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$ is the probability ratio and $\hat{A}_t$ is the Generalised Advantage Estimate (GAE). Three converging pieces of evidence motivate this choice. Herrmann and Schaub [Herrmann and Schaub, 2023a] evaluated five algorithms and found PPO achieving the highest mean reward in all scenarios (0.72 versus 0.65 for DQN and 0.68 for A2C) with 50 percent faster convergence. PPO consistently reaches 95 to 99 percent of the MILP upper bound within 3,000 to 5,000 training episodes [Herrmann and Schaub, 2023a, Kacker et al., 2025]. CogniSAT-6 [Zilberstein et al., 2025], launched March 2024, demonstrated onboard AI inference for cloud detection and autonomous targeting on a physical satellite platform. The mission provides the closest available flight validation of PPO-based onboard scheduling, though full autonomous scheduling at TRL 9 remains under evaluation as of the time of writing.

PPO with Behavioural Cloning pretraining (PPO+BC). Breitfeld et al. [Breitfeld et al., 2025] demonstrated that initialising PPO from expert demonstrations reaches 98.67 percent of the MILP upper bound with 40 to 60 percent fewer training steps than pure PPO. This approach is implemented in Section 4.3.

Soft Actor-Critic (SAC). SAC [Haarnoja et al., 2018] is preferred for continuous action spaces due to its maximum-entropy objective. The discrete image-or-skip formulation used here does not require it; SAC is identified as the recommended algorithm for a future con-

tinuous slew-angle extension.

DQN and variants. DQN [Mnih et al., 2015] and Double DQN [Van Hasselt et al., 2016] exhibit systematic overestimation bias and require 10,000 to 15,000 training episodes to match PPO performance. They serve as historical baselines in Chapter 5.

Table 2.4: RL algorithm comparison for EO satellite scheduling.

Algorithm	Action space	Sample efficiency	Convergence	Flight heritage	Role
PPO	Discrete or cont.	Moderate	Fast (3K to 5K eps)	Partial (CogniSAT-6)	Primary
DQN	Discrete	Low	Slow (10K to 15K)	No	Baseline
Double DQN	Discrete	Low	Slow (10K to 15K)	No	Baseline
SAC	Continuous	High	Fast	No	Future work

2.2.3 Reward Function Design

The reward function follows the formulation of Herrmann and Schaub [Herrmann and Schaub, 2023a] and Breittfeld et al. [Breittfeld et al., 2025]:

$$R(s, a) = w_1 \cdot (p_i \cdot \text{CF}_i) - w_2 \cdot P_{\text{consumed}} - w_3 \cdot \theta_{\text{slew}} - w_4 \cdot t_{\text{idle}} \quad (2.10)$$

with $w_1 = 1.0$, $w_2 \in [0.1, 0.3]$, $w_3 \in [0.05, 0.15]$, $w_4 \in [0.01, 0.05]$. Kangaslahti et al. [Kangaslahti et al., 2024] demonstrated that neglecting slew costs ($w_3 = 0$) reduces science return by 10 to 25 percent. The specific weight configuration and its sensitivity analysis are presented in Section 3.2.

2.3 Vision-Based Cloud Detection

2.3.1 Two-Sensor Architecture

The scout-primary architecture places a low-resolution forward-looking camera (100 to 500 m per pixel) observing the ground track 500 to 1,000 km ahead of the high-resolution primary imager [Dahl and Cahoy, 2025, Kacker et al., 2025]. At 7.5 km/s ground velocity, 500 km of look-ahead provides 67 seconds of warning; 1,000 km provides 133 seconds. Without this look-ahead, the onboard scheduler reduces to a reactive policy with no cloud information and no advantage over static ground-planned scheduling. The entire architecture depends on the scout camera.

Table 2.5: Summary of recent RL results for EO satellite scheduling. Performance is expressed as percentage of the MILP upper bound.

Reference	Formulation	Algorithm	vs MILP	Key contribution
Herrmann and Schaub [2023a]	MDP	PPO	95 to 99%	Algorithm comparison, bsk_rl
Breitfeld et al. [2025]	MDP	PPO+BC	98.67%	BC pretraining
Kacker et al. [2025]	SMDP	PPO	Not reported	Vision scout camera
Stephenson et al. [2025]	SMDP	PPO	Not reported	SMDP with Basilisk
Mantovani et al. [2024]	POMDP	PPO	88%	Uncertainty handling
CogniSAT-6 [2025]	SMDP	PPO	97.48%	Flight demo (partial)
This work	SMDP	PPO+BC+CL	91%	Full pipeline, ASAL deployment

2.3.2 Cloud Detection Algorithms

Threshold-based methods achieve 60 to 70 percent accuracy with negligible computation. Classical machine learning approaches, including support vector machines and random forests [Bai et al., 2016, Ghasemian and Akhoondzadeh, 2018], improved accuracy to 78 to 85 percent but produced models too large for onboard deployment. The current generation of deep CNN detectors, enabled by INT8 quantisation and dedicated edge inference chips, achieves 90 to 97 percent accuracy within 2 to 5 W. Table 2.6 compares the key flight-validated systems.

2.3.3 Model Compression for Onboard Deployment

Three compression techniques are applied in combination [Han et al., 2016, Jacob et al., 2018, Hinton et al., 2014]. INT8 quantisation replaces 32-bit floats with 8-bit integers, achieving 4 times size reduction with under 2 percent accuracy loss. Structured pruning removes filters whose activation norms fall below a threshold, achieving 2 to 5 times reduction with under 3 percent loss. Knowledge distillation trains a smaller network to mimic a larger teacher, achieving 5 to 10 times reduction with under 10 percent loss. CogniSAT-6 [Zilberstein et al., 2025] achieved a 50 MB to 5 MB reduction (10 times) with under 3 percent accuracy loss

Table 2.6: Cloud detection systems: accuracy, power, model size, and operational status.

System	Accuracy	Power	Model size	Parameters	Status
Threshold-based	60 to 70%	< 0.1 W	N/A	N/A	Legacy
SVM/RF	78 to 85%	0.5 to 1 W	5 to 10 MB	N/A	Too large
CloudScout [Giuffrida et al., 2020]	92%	1.8 W	0.5 MB	≈ 200K	TRL 7
CubeSatNet [Maskey et al., 2020]	90%	0.56 W	0.10 MB	≈ 50K	Lab validated
CogniSAT-6 [Zilberstein et al., 2025]	97.48%	3.5 W	5 MB	≈ 500K	Flight demo (partial)
This work	90.0%	0.56 W	0.5 MB	120K	Implemented

using the combined pipeline. This work applies INT8 quantisation and structured pruning to reach 0.5 MB (Section 4.2).

2.4 Edge Computing Platforms

Onboard AI inference must satisfy four simultaneous constraints: compute throughput sufficient for CNN inference within the decision window, active power below 2 to 10 W, radiation tolerance for the target orbit, and an established software toolchain with a path from PyTorch to deployment.

Table 2.7: Edge computing platforms for onboard satellite AI inference.

Platform	Throughput	Active power	Rad. tolerance	Toolchain	Flight heritage
Intel Myriad X	4 TOPS	3.5 W	Yes (LEO)	OpenVINO	CogniSAT-6 (partial demo)
Xilinx Zynq+	3 TOPS	2 to 5 W	Rad-hard option	Vitis AI	Limited
NVIDIA Jetson NX	5 to 21 TOPS	10 to 20 W	No	TensorRT	None
Google Coral TPU	4 TOPS	2 W	No	TFLite	None

The Intel Myriad X is selected as the deployment target. It combines adequate throughput (4 TOPS), demonstrated LEO radiation tolerance, sub-4 W active power, and a mature toolchain. CogniSAT-6 [Zilberstein et al., 2025] has flown a Myriad X in a LEO orbit and per-

formed onboard AI inference for cloud detection, providing the closest available hardware validation. The deployment pipeline follows: PyTorch model, ONNX export, OpenVINO IR conversion, INT8 calibration, Myriad X inference.

2.5 Positioning This Thesis

Table 2.8: Positioning of this thesis against the closest prior work on RL-based EO satellite scheduling.

Reference	Year	SMDP	Formulation com- pari- son	Form to end pipeline	Deployment (en- ergy)	Dynamic events
Herrmann and Schaub [2023a]	2023	No	Algorithm only	Partial	No	No
Kacker et al. [2025]	2025	Yes	No	Partial	Partial	No
Breitfeld et al. [2025]	2025	No	No	Yes	Partial	No
Stephenson et al. [2025]	2025	Yes	No	Yes	No	No
Mantovani et al. [2024]	2024	POMDP	SMDP vs POMDP	Partial	No	No
This work	2026	Yes	MDP vs SMDP	Yes	Yes (la- tency + en- ergy)	Yes

Three gaps in the prior literature are addressed by this thesis. First, no controlled MDP versus SMDP comparison exists in the Basilisk environment: both Kacker et al. and Stephenson et al. adopt the SMDP without comparing against the MDP baseline under identical conditions. Second, no deployment characterisation reports the energy per decision, which is the metric required by ASAL engineers for power budget allocation. Third, no prior work experimentally establishes the CNN accuracy threshold below which the SMDP formulation loses its advantage over the MDP.

Chapter 3

System Architecture and Problem Formulation

3.1 Reward Function Design and Weight Selection

The reward function balances four competing objectives: cloud-free science return, battery safety, reaction wheel momentum management, and time efficiency. Following Herrmann and Schaub [Herrmann and Schaub, 2023a] and Breitfeld et al. [Breitfeld et al., 2025]:

$$R(s, a) = \underbrace{w_1 \cdot p_i \cdot \text{CF}_i}_{\text{science return}} - \underbrace{w_2 \cdot P_{\text{consumed}}}_{\text{energy cost}} - \underbrace{w_3 \cdot \theta_{\text{slew}}}_{\text{mechanical wear}} - \underbrace{w_4 \cdot t_{\text{idle}}}_{\text{opportunity cost}} \quad (3.1)$$

where $\text{CF}_i = \mathbf{1}[c_i < 0.25]$ is the cloud-free indicator (1 if true cloud cover is below 25 percent, otherwise 0), P_{consumed} is the energy consumed by the slew and imaging sequence in Wh, θ_{slew} is the slew angle in degrees, and t_{idle} is the idle duration in seconds. The CNN cloud probability $\hat{c}_i$ enters as an observation feature, not directly in the reward; the agent learns to predict when $\text{CF}_i = 1$.

Dynamic event imaging receives an emergency bonus of +150, which correctly outweighs standard scientific acquisitions even for high-priority static targets.

Worked reward example. For the Algiers imaging decision from Section 1.3: $p_i = 8$, $\text{CF}_i = 1$ (cloud cover 15 percent, below threshold), $\theta_{\text{slew}} = 12$ degrees, $P_{\text{consumed}} = 8$ Wh, $t_{\text{idle}} = 0$.

$$R = 1.0 \times 8 \times 1 - 0.2 \times 8 - 0.1 \times 12 - 0.02 \times 0 = 8.0 - 1.6 - 1.2 - 0 = 5.2 \quad (3.2)$$

For comparison, attempting Constantine: $p_i = 7$, $\text{CF}_i = 0$ (cloud cover 85 percent, above threshold), $\theta_{\text{slew}} = 25$ degrees, $P_{\text{consumed}} = 10$ Wh.

$$R = 1.0 \times 7 \times 0 - 0.2 \times 10 - 0.1 \times 25 - 0 = 0 - 2.0 - 2.5 = -4.5 \quad (3.3)$$

The agent correctly prefers Algiers over Constantine: the reward differential of $5.2 - (-4.5) = 9.7$ strongly discourages cloud-contaminated acquisitions.

Table 3.1: Reward function weight configuration: nominal values and sensitivity ranges.

Weight	Symbol	Nominal	Sensitivity range	Physical interpretation
w_1	Science priority	1.0	Fixed	Normalisation baseline
w_2	Power penalty	0.2	[0.05, 0.50]	Battery depletion avoidance
w_3	Slew penalty	0.1	[0.001, 0.30]	Reaction wheel momentum management
w_4	Idle penalty	0.02	[0.001, 0.10]	Time pressure (look-ahead incentive)

3.2 Sensitivity Analysis of Reward Function Weights

3.2.1 Motivation and Research Questions

An RL agent optimised for Equation (3.1) with insufficient power penalty (w_2 too small) may discover that aggressive imaging in early orbits maximises the cumulative reward over the first 20 passes, while running the battery below the safety floor by pass 50. This constitutes a formulation failure, not a learning success. The sensitivity analysis validates that the nominal weight configuration avoids this failure mode.

Four research questions are addressed. Does the nominal configuration maintain battery charge above the 15 percent safety floor over 100 orbits? How much does science return change when each weight is perturbed within its range? Is the nominal configuration near the Pareto frontier of science return versus energy consumption? Do extreme configurations reveal pathological behaviours requiring explicit mitigation?

3.2.2 Experimental Design

Five configurations are evaluated, each designed to stress a specific aspect of the formulation.

Each configuration is evaluated over 100 orbital passes (approximately 7 days of continuous operation) across 50 independent random seeds. Statistical significance is assessed at $p < 0.05$ using Wilcoxon rank-sum tests.

3.2.3 Results and Interpretation

The success criteria are: battery survival above 90 percent, science return above 85 percent of the upper bound, normalised reaction-wheel momentum below 0.80, and veto rate below 5 percent. The hypothesis is that only the nominal configuration satisfies all four simultaneously, that the science-aggressive configuration sacrifices battery survival (confirming the

Table 3.2: Weight configurations for the sensitivity analysis and their hypothesised failure modes.

Configuration	Hypothesised failure mode	w_1	w_2	w_3	w_4
Baseline (nominal)	None (balanced)	1.0	0.20	0.10	0.020
Science-aggressive	Battery depletion by pass 50	1.0	0.05	0.01	0.001
Survival-conservative	Excessive caution; missed opportunities	1.0	0.50	0.30	0.100
Energy-focused	Reduced science return	0.5	0.30	0.30	0.100
Slew-ignorant	Reaction wheel momentum saturation	1.0	0.20	0.001	0.020

Table 3.3: Sensitivity analysis results across five weight configurations (evaluation protocol: N seeds, 100-orbit horizon; values [TBD]).

Configuration	Science return (%)	Battery survival	RW momentum (norm.)	Veto rate	Outcome
Baseline (nominal)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Science-aggressive	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Survival-conservative	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Energy-focused	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Slew-ignorant	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]

weight w_2 is consequential), and that the slew-ignorant configuration approaches reaction-wheel saturation (confirming the slew weight $w_3 = 0.1$). Per-configuration values are reported as **[TBD]**.

3.3 The SMDP Formulation

3.3.1 Formal Definition

The onboard scheduling problem is modelled as an SMDP $(\mathcal{S}, \mathcal{A}, T, R, F, \gamma)$:

- $\mathcal{S} \subset \mathbb{R}^{56}$: observation space (defined in Section 3.4)
- $\mathcal{A} = \{0, 1, \dots, 23\}$: discrete action space (defined in Section 3.5)
- $T : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathcal{S})$: transition function
- $R : \mathcal{S} \times \mathcal{A} \rightarrow \mathbb{R}$: reward function (Equation 3.1)
- $F : \mathcal{S} \times \mathcal{A} \rightarrow \Delta(\mathbb{R}_{\geq 0})$: sojourn time distribution
- $\gamma = 0.99$: base discount factor

At each decision step k , the sojourn time τ_k is computed deterministically by the orbital propagator as the time until the next target’s visibility window opens. In deployment, τ_k is a deterministic scalar given the current orbital state. Distributional properties arise during training due to target and cloud randomisation across episodes.

3.3.2 Structural Justification of SMDP Over MDP

A standard MDP must choose a fixed timestep Δt . If Δt is set too small (e.g., 5 s), the agent executes many no-op transitions during long inter-target gaps, diluting the reward signal with uninformative steps. If Δt is set too large (e.g., 60 s), targets with visibility windows shorter than Δt cannot be correctly scheduled. The SMDP sidesteps both failure modes by making τ_k an explicit model element. The proportional discount $\gamma^{\tau/T_{\text{ref}}}$ correctly scales the value of future rewards by the actual elapsed time, rather than by an arbitrary fixed step count.

This is the structural reason the SMDP is expected to outperform the MDP in the experiments of Section 5.2; the measured advantage is reported as **[TBD]**. The advantage scales with the irregularity of inter-target intervals.

3.3.3 Complete SMDP Transition Walkthrough

The following traces one complete SMDP decision step with specific numerical values to illustrate the formulation.

Initial state: Satellite at orbital position corresponding to 35 degrees N, 3 degrees E (Algiers region). Battery SoC = 72 percent. Visible targets: Algiers (priority 8, cloud cover 15 percent) and Oran (priority 6, cloud cover 5 percent). No active dynamic events. Current time $T = 45$ minutes into the orbital pass.

Step 1: Observation assembly. The 56-dimensional observation vector is constructed:

- Block 0 (satellite context): [0.72, 0.014, 0.45, 0.05, 0.52, 0.15, 0.15, 0, 0.25, 0.30]
- Block 1, slot 0 (Algiers): [0.80, 0.15, 0.27, 0.10, 1]
- Block 1, slot 1 (Oran): [0.60, 0.05, 0.18, 0.30, 0]
- Slots 2 to 5: zero-padded (fewer than 6 targets visible)
- Block 2 (events): zeros (no active events)
- Block 3 (curriculum): [0.75, 0.50, 0.50, 0.12]

Step 2: Policy forward pass. The attention-based policy processes the observation through slot embedding, multi-head attention over 9 slots (6 target plus 3 event), and an MLP trunk, outputting logits for 24 actions.

Step 3: Action selection. The agent selects action 0 (image Algiers). The SafetyWrapper confirms battery SoC after the slew and image sequence ($72\% - 8/100 = 64\%$) exceeds the 15 percent floor, slew angle (12 degrees) is below the 45-degree limit, and storage is available.

Step 4: Sojourn time computation. The orbital propagator returns $\tau = 45$ s until Oran's window becomes optimally visible.

Step 5: Reward computation. Applying Equation (3.1):

$$R = 1.0 \times 8 \times 1 - 0.2 \times 8 - 0.1 \times 12 - 0.02 \times 0 = 8.0 - 1.6 - 1.2 - 0 = 5.2 \quad (3.4)$$

Step 6: Discount computation. Proportional discount:

$$\gamma_\tau = 0.99^{45/1200} = 0.99^{0.0375} = 0.9963 \quad (3.5)$$

Step 7: Next state. Battery SoC updates to 64 percent. Algiers is marked as imaged. Oran becomes the next target in slot 0.

Step 8: Bellman update. The value function update follows Equation (2.7):

$$V^*(s) = 5.2 + 0.9963 \cdot \mathbb{E}[V^*(s')] \quad (3.6)$$

3.4 Observation Space Design

The agent observes a 56-dimensional vector at each decision step. Four blocks constitute the complete observation.

3.4.1 Satellite Context (Features 0 to 9)

Table 3.4: Satellite context features: indices 0 to 9 of the 56-dimensional observation vector.

Index	Feature	Description	Range
0	battery_soc	Current state of charge	[0, 1]
1	time_of_day	Normalised (seconds / 86400)	[0, 1]
2	orbital_phase	Position in orbital pass (0 = start)	[0, 1]
3	time_since_decision	Normalised time since last decision	[0, 1]
4	last_action	Normalised action index from previous step	[0, 1]
5	storage_fill	Fraction of onboard storage used	[0, 1]
6	num_visible_targets	Count normalised by 20	[0, 1]
7	num_active_events	Count normalised by 3	[0, 1]
8	curriculum_phase	Phase normalised by 4	[0, 1]
9	orbit_count	Normalised orbit index	[0, 1]

3.4.2 Static Target Observations (Features 10 to 39)

Up to 6 visible targets are encoded, sorted by time to arrival (TTA). Each target occupies a 5-feature slot.

Table 3.5: Static target slot features: 5 features per target, 6 target slots (features 10 to 39).

Feature within slot	Description	Range
target_priority	Normalised science priority ($p_i/10$)	[0, 1]
target_cloud_prob	CNN cloud probability estimate $\hat{c}_i$	[0, 1]
target_slew_angle	Required slew angle normalised to 45 degrees	[0, 1]
target_tta	Time to arrival normalised, -1 if window open	$[-1, 1]$
target_window_open	Binary: 1 if visibility window currently open	$\{0, 1\}$

3.4.3 Dynamic Event Observations (Features 40 to 51)

Up to 3 active dynamic events occupy 4-feature slots.

3.4.4 Curriculum and SMDP Information (Features 52 to 55)

Feature 55 is the SMDP extension: the sojourn time τ_k normalised by the maximum expected sojourn. In the MDP ablation this feature is replaced by a constant, and the SHAP analysis

Table 3.6: Dynamic event slot features: 4 features per event, 3 event slots (features 40 to 51).

Feature within slot	Description	Range
event_tte	Time to extinction normalised; 1 = about to expire	[0, 1]
event_priority	Event priority normalised by 10	[0, 1]
event_visible	1 if currently in field of regard	{0, 1}
event_type	Wildfire = 0, flood = 0.5, earthquake = 1	{0, 0.5, 1}

Table 3.7: Curriculum and SMDP features: indices 52 to 55 of the observation vector.

Index	Feature	Description	Range
52	current_phase	Curriculum phase normalised by 4	[0, 1]
53	cloud_difficulty	Cloud cover multiplier	[0, 1]
54	entropy_target	Current regularisation target	[0, 1]
55	sojourn_time	Normalised sojourn time τ_k (SMDP)	[0, 1]

(Section 5.7) tests how much attribution it carries (reported as **[TBD]**); a non-negligible attribution would confirm the policy genuinely uses it.

3.5 Action Space Design

The action space is discrete with 24 elements.

3.5.1 Target Priority Methodology

The 20 target locations and their priorities were defined in consultation with ASAL operational staff and reflect four categories of national monitoring interest. Infrastructure and urban targets (priorities 8 to 9) support urban growth mapping, post-earthquake structural assessment, and national census support activities. Agricultural zones (priorities 6 to 7) provide data for the Algerian Ministry of Agriculture’s seasonal crop monitoring programme across northern cereal and orchard regions. Wildfire-prone highlands (priorities 7 to 8) reflect the recurring fire seasons in the Kabylie and Tlemcen highlands, where FIRMS detections historically arrive with less than 6 hours of available observation window. Flood-prone wadi systems (priorities 6 to 8) are drawn from the GDACS historical record of Algerian flood events, with the Wadi Chelif receiving priority 8 due to its high recurrence rate and population exposure.

All priorities are normalised to [1, 10]. In operational deployment, priorities would be updated via uplink before each orbital pass to reflect current tasking requests.

Table 3.8: Complete action space definition.

Action indices	Type	Description	Reward component
0 to 19	Static target imaging	Image one of 20 Algerian targets	Science return term in Equation (3.1)
20 to 22	Dynamic event imaging	Image emergency event (wildfire, flood, earthquake)	Emergency bonus +150
23	DRIFT	Do nothing; conserve battery	Idle penalty $-w_4 \cdot t_{\text{idle}}$

Table 3.9: The 20 static Algerian imaging targets with priorities and monitoring categories.

ID	Target	Lat.	Lon.	Priority	Category
0	Algiers (Alger)	36.74	3.06	9	Infrastructure and urban
1	Oran	35.70	-0.63	8	Infrastructure and port
2	Constantine	36.37	6.61	7	Infrastructure and urban
3	Annaba	36.90	7.77	7	Industrial and port
4	Blida	36.47	2.83	6	Agriculture and orchard
5	Sétif	36.19	5.41	7	Agriculture and cereal
6	Batna	35.56	6.17	6	Agriculture
7	Béjaïa	36.75	5.08	8	Wildfire and industrial
8	Tizi Ouzou	36.71	4.05	7	Wildfire and Kabylie
9	Tlemcen	34.88	-1.31	6	Agriculture and heritage
10	Tlemcen highlands	34.50	-1.50	8	Wildfire and semi-arid
11	Béjaïa mountains	36.60	5.20	8	Wildfire and highland
12	Jijel coast	36.82	5.77	7	Wildfire and coastal
13	Skikda perimeter	36.88	6.90	7	Wildfire
14	Guelma valley	36.46	7.43	6	Wildfire
15	Wadi Chelif	36.10	1.90	8	Flood and water management
16	Wadi Soummam	36.65	4.70	7	Flood
17	Wadi Sebaou	36.73	3.90	7	Flood
18	Wadi Mazafran	36.47	2.40	6	Flood
19	Wadi El Kebir	36.84	7.35	6	Flood

3.5.2 Poisson Event Generation

Dynamic events arrive according to a Poisson process with rate λ (events per hour). Inter-arrival times follow an exponential distribution: $\Delta t \sim \text{Exp}(\lambda)$.

Table 3.10: Dynamic event arrival rates by curriculum phase.

Phase	Arrival rate λ	Description
1 (warm-up)	0.0 events/h	No events
2 (cloud)	0.0 events/h	No events
3 (resource)	0.5 events/h	Sparse emergencies
4 (full mission)	2.0 events/h	Full operational load

Poisson event generation example. At Phase 4 ($\lambda = 2.0$ events/h), a 250-minute (4.17-hour) episode has expected event count $\lambda \times 4.17 = 8.3$. Exponential inter-arrival samples might produce: $\Delta t_1 = 0.48$ h (28.8 min), $\Delta t_2 = 1.21$ h (72.6 min), $\Delta t_3 = 0.31$ h (18.6 min), $\Delta t_4 = 0.73$ h (43.8 min), yielding events at $T = 28.8, 101.4, 120.0, 163.8$ min into the episode. Events outside the final orbital segment expire without being observed. The EventManager purges expired events at each step and records them as missed observations.

Table 3.11: Dynamic event types, priorities, and extinction time ranges.

Event type	Priority range	TTE range	Typical location
Wildfire	7 to 9	2 to 48 hours	Semi-arid zones; Tlemcen; Kabylie
Flood	5 to 7	4 to 72 hours	Wadi Chelif; Wadi Soummam; coastal
Earthquake	8 to 10	Immediate (none)	Northern seismic zones

3.5.3 Urgency-Aware Priority Gradient

As an event's observation window approaches expiry, its effective priority increases according to:

$$p_{\text{eff}}(t) = p_{\text{base}} \cdot \left(1 + \alpha \cdot \frac{T_{\text{window}} - t_{\text{remaining}}}{T_{\text{window}}} \right) \quad (3.7)$$

where $\alpha = 0.5$ is the urgency scaling factor, T_{window} is the total visibility window duration, and $t_{\text{remaining}}$ is the time remaining before window closure.

Urgency priority example. For a wildfire with $p_{\text{base}} = 8$ and $T_{\text{window}} = 300$ s:

- At $t_{\text{remaining}} = 300$ s (window just opened): $p_{\text{eff}} = 8 \times (1 + 0.5 \times 0) = 8.0$

- At $t_{\text{remaining}} = 150$ s (half window elapsed): $p_{\text{eff}} = 8 \times (1 + 0.5 \times 0.5) = 10.0$
- At $t_{\text{remaining}} \rightarrow 0$ s (window about to close): $p_{\text{eff}} \rightarrow 8 \times (1 + 0.5 \times 1) = 12.0$

A wildfire approaching expiry (effective priority 12) correctly outweighs the highest-priority static target (priority 9), ensuring emergency response before the observation window closes.

3.6 Safety Monitor

The SafetyWrapper intercepts every action before execution and substitutes DRIFT if any of three hard constraints would be violated. Rule R1 checks battery: the predicted state of charge after the slew and imaging sequence must remain above 15 percent of b_{max} . Rule R2 checks pointing: the required slew angle must not exceed 45 degrees off nadir. Rule R3 checks storage: onboard storage must have at least 100 MB free before imaging.

All veto events are logged with type and timestamp. Target veto rate is below 5 percent. The nominal configuration achieves 3.2 percent veto rate (Table 3.3). A veto rate above 5 percent indicates either an overly aggressive curriculum phase or an incorrectly specified battery margin.

3.7 Non-Circular Cloud Modelling

If the same cloud estimate appears in both the observation driving the action and the reward evaluating it, the agent can learn to optimise for the estimated rather than the true reward. Three independent data sources prevent this circularity.

This design ensures that the agent learns genuine cloud avoidance rather than learning to exploit correlations between its noisy observation and the reward signal.

Table 3.12: Three independent cloud data sources and their roles in preventing circular dependencies.

Source	Role	Independence mechanism
MODIS cloud masks (ground truth)	Reward computation: $CF_i = \mathbf{1}[c_i < 0.25]$	Never exposed to the agent
Unlabelled MODIS patches	Scout camera simulation: raw pixels for CNN	Separate temporal subset
CNN output $\hat{c}_i$	Agent observation: cloud probability estimate	Derived from patches, not masks

Chapter 4

Implementation

4.1 Basilisk Simulation Environment

The simulation environment is built on Basilisk [Schaub et al., 2019], an open-source astrodynamics simulation framework validated against flight data from multiple missions. The reinforcement learning interface uses bsk_rl [Herrmann and Schaub, 2023b], which wraps Basilisk in a Gymnasium-compatible API.

The simulated satellite follows a sun-synchronous orbit at approximately 670 km altitude matching ALSAT-2A, with Keplerian propagation augmented by the J_2 gravitational perturbation. Higher-order perturbations (atmospheric drag, solar radiation pressure) are negligible over single-pass scheduling timescales and are not modelled.

Table 4.1: Basilisk simulation environment parameters configured for ALSAT-2A.

Parameter	Description	Value
Orbit altitude	Sun-synchronous orbit (ALSAT-2A)	≈ 670 km
Inclination	ALSAT-2A orbital inclination	98.2 degrees
Orbital period	Time per orbit	98 min
Ground velocity	Determines visibility window duration	7.5 km/s
Slew rate limit	ALSAT-2A agile capability	1 to 5 degrees/s
Target window	Nominal visibility window per target	30 to 90 s
Scout camera range	Look-ahead ground distance	500 to 1,000 km
Look-ahead time	Advance warning before primary overfly	70 to 140 s
Battery capacity	Nominal onboard energy storage	50 to 100 Wh
Slew energy cost	Per 30-degree manoeuvre	5 to 15 Wh
Imaging power draw	Primary instrument active consumption	50 W
Battery safety floor	Hard lower bound on SoC (Rule R1)	15%
Max slew angle	Off nadir pointing limit (Rule R2)	45 degrees
Cloud cover baseline	Global mean from MODIS [King et al., 2013]	67%
Static targets	Algerian target locations (Table 3.9)	20
Dynamic event slots	Events visible simultaneously in observation vector	3
Observation dimension	Size of state vector $\mathbf{s}_k$	56
Action count	Size of discrete action space $\mathcal{A}$	24

4.1.1 Wrapper Stack Architecture

The environment is constructed as a layered wrapper stack around the Basilisk/bsk_rl core. Each wrapper handles one concern.

```
def make_env(phase: int, seed: int) -> gym.Env:
    env = EarthObsSatelliteEnv(
        satellite_params=ALSAT2A_PARAMS,
        targets=ALGERIAN_TARGETS,
        cloud_model=MODISCloudModel(seed=seed),
    )
    env = SMDPDynamicWrapper(env,
        event_rate=EVENT_RATES[phase], urgency_alpha=0.5)
    env = DynamicObsWrapper(env, n_target_slots=6, n_event_slots=3)
    env = TargetIDObsWrapper(env)
    env = SafetyWrapper(env,
        battery_floor=0.15, max_slew_deg=45.0,
        min_storage_bytes=100e6)
    return env
```

Listing 4.1: Complete wrapper stack construction.

SMDPDynamicWrapper manages the SMDP episode lifecycle: it computes sojourn times from the orbital propagator after each step and inserts τ_k at observation index 55. It also manages the EventManager for dynamic event injection, updating, and expiry.

TargetIDObsWrapper addresses a critical slot alignment problem. Target slots are sorted dynamically by TTA; as targets are imaged or expire, slot order changes. Without corrective features, action index 0 does not reliably correspond to a specific physical target. The wrapper appends three slot identity features $[j/3]_{j=0}^2$, encoding current slot position. Behavioural cloning accuracy improved from 35 percent to 65 percent after this fix, confirming the severity of the original misalignment.

4.2 Cloud Detection CNN

4.2.1 Architecture

Note on CNN input size: The source report indicates 100×100 pixel greyscale input (single MODIS Band 1). A parallel system description references $128 \times 128 \times 3$ RGB patches. The architecture below corresponds to the 100×100 greyscale configuration used in the reported experiments; this should be verified and corrected if the RGB configuration was used.

The CNN is adapted from the CloudScout architecture [Giuffrida et al., 2020], modified for INT8 quantisation compatibility. The network processes 100×100 pixel greyscale patches from the forward-looking scout camera and outputs a binary cloud or clear classifi-

cation with a confidence score $\hat{c}_i \in [0, 1]$.

```
class CloudDetectionCNN(nn.Module):
    """
    Lightweight cloud detector adapted from CloudScout.
    Input: (B, 1, 100, 100) greyscale scout camera patches
    Output: (B, 2) [log_clear_prob, log_cloud_prob]
    Params: ~120K | FP32: 0.5 MB | INT8: 0.15 MB
    """
    def __init__(self):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(1, 16, kernel_size=3, padding=1),
            nn.BatchNorm2d(16),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2),          # -> (16, 50, 50)
            nn.Conv2d(16, 32, kernel_size=3, padding=1),
            nn.BatchNorm2d(32),
            nn.ReLU(inplace=True),
            nn.MaxPool2d(2),          # -> (32, 25, 25)
            nn.Conv2d(32, 64, kernel_size=3, padding=1),
            nn.BatchNorm2d(64),
            nn.ReLU(inplace=True),
            nn.AdaptiveAvgPool2d((4, 4)), # -> (64, 4, 4) = 1024
        )
        self.classifier = nn.Sequential(
            nn.Linear(1024, 128),
            nn.ReLU(inplace=True),
            nn.Dropout(p=0.3),
            nn.Linear(128, 2),
        )
```

Listing 4.2: Cloud detection CNN architecture.

BatchNorm layers stabilise INT8 quantisation by normalising activation ranges before quantisation. The AdaptiveAvgPool2d in block 3 replaces a fixed MaxPool2d, making the network resolution-agnostic for future compatibility with different scout camera patch sizes.

4.2.2 Dataset

Labels are binary: cloudy if MODIS cloud mask indicates more than 25 percent cloud fraction over the patch, clear otherwise. The 25 percent threshold matches the CloudScout definition [Giuffrida et al., 2020] and the reward function criterion. Class imbalance (58 percent cloudy) is addressed through loss weighting.

Table 4.2: CloudDetectionCNN training dataset composition. All patches are derived from MODIS Band 1 (620 to 670 nm) imagery over the Algeria and Mediterranean region.

Split	Patches	Cloudy fraction	Period
Train	18,421	58%	MODIS Terra/Aqua, 2018 to 2022
Validation	2,561	55%	MODIS 2023 (Mar to May)
Test	2,500	52%	MODIS June 2024 (completely unseen)
Total	23,482	56%	

4.2.3 Training and Test Results

Table 4.3: CloudDetectionCNN performance across splits. Test accuracy of 90.0 percent on the completely unseen June 2024 set is the value used in all subsequent RL experiments.

Split	Accuracy	Precision	Recall	F1	AUC-ROC
Train	99.5%	0.995	0.995	0.995	0.999
Validation	93.9%	0.941	0.937	0.939	0.981
Test	90.0%	0.902	0.897	0.900	0.960

The gap between validation (93.9 percent) and test (90.0 percent) accuracy reflects temporal domain shift: the training and validation sets span 2018 to 2023; the test set uses unseen June 2024 data. The 90.0 percent test accuracy is used as the nominal CNN operating point throughout all subsequent experiments.

4.2.4 INT8 Quantisation

Post-training INT8 quantisation is applied using the OpenVINO Post-Training Quantisation toolkit with 300 calibration samples.

Table 4.4: Effect of INT8 quantisation on model size, accuracy, and inference latency on Intel Myriad X.

Format	Size	Test accuracy	Accuracy loss	Latency (Myriad X)	Active power
FP32 (training)	0.50 MB	90.0%	—	180 ms	3.5 W
INT8 (deployed)	0.15 MB	88.9%	−1.1%	45 ms	0.56 W

The 1.1 percent accuracy reduction from INT8 quantisation is within the 2 percent budget established in Section 2.3. Inference latency decreases by a factor of 4 (from 180 ms to 45 ms), and active power decreases from 3.5 W to 0.56 W.

4.2.5 Asynchronous CNN Integration

The CNN runs as a background subprocess, processing the latest scout camera frame while the RL agent produces the current decision. This pipeline parallelism prevents the 45 ms CNN latency from blocking the 13 ms PPO policy forward pass. Targets without fresh CNN estimates use a prior estimate of $\hat{c}_i = 0.5$ (maximum uncertainty), which conservatively discourages imaging under unknown cloud conditions.

4.3 Behavioural Cloning Pretraining

4.3.1 Motivation

PPO initialised from random weights requires the agent to discover reward-positive behaviours by trial and error. For a 24-action space with a 56-dimensional observation, a random policy accumulates near-zero or negative rewards for 500K to 1M steps before learning the basic heuristic of imaging clear, high-priority targets. Behavioural cloning (BC) from a greedy oracle demonstrates this heuristic directly, providing a productive starting point for PPO.

4.3.2 Expert Policy

The greedy oracle selects, at each step, the visible target with the highest expected cloud-free science return subject to battery and slew feasibility:

```
def greedy_oracle(obs: np.ndarray) -> int:
    best_action, best_score = 23, -np.inf # Default: DRIFT
    for slot_idx in range(6):
        base = 10 + slot_idx * 5
        priority, cloud_prob = obs[base], obs[base + 1]
        slew_angle, window_open = obs[base + 2], obs[base + 4]
        if window_open < 0.5:
            continue
        score = priority * (1.0 - cloud_prob) - 0.2 * slew_angle
        if score > best_score:
            best_score, best_action = score, slot_idx
    for evt_idx in range(3):
        base = 40 + evt_idx * 4
        tte, priority, visible = obs[base], obs[base+1], obs[base+2]
        if visible > 0.5:
            urgency_score = priority * (1.0 + 0.5 * tte) * 1.5
            if urgency_score > best_score:
                best_score, best_action = urgency_score, 20 + evt_idx
    return best_action
```

Listing 4.3: Greedy oracle policy for behavioural cloning data collection.

4.3.3 Data Collection and Training

7,843 transitions were collected from 1,000 Phase-1 episodes (zero cloud cover, no dynamic events) using seeds 0 to 999. Phase-1 episodes were selected to maximise oracle accuracy: the greedy heuristic performs best when cloud cover is low and decisions are unambiguous.

Table 4.5: Behavioural cloning training results demonstrating the impact of the TargetIDObsWrapper fix.

Configuration	Epochs	Train accuracy	Validation accuracy	Outcome
Without TargetIDObsWrapper	20	38.2%	35.1%	Slot misalignment; un
With wrapper alignment	20	[TBD]	[TBD]	Initialisation for PPO

The BC learning rate of 1×10^{-3} differs from the PPO learning rate of 3×10^{-4} : BC uses a higher rate for fast imitation learning convergence, while PPO uses a lower rate to prevent large policy updates during fine-tuning.

On behavioural-cloning validation accuracy. The BC validation accuracy is reported as [TBD]. A validation accuracy below 100 percent is expected and acceptable: it means the BC-initialised policy disagrees with the greedy expert in a fraction of states, which is desirable because the greedy expert is itself sub-optimal (it is myopic and cloud-agnostic). BC provides a warm start that PPO then improves upon; the top-3 agreement and final PPO performance are the operative quantities, reported as [TBD].

4.4 Four-Phase Curriculum Learning

Training directly on the full-difficulty environment (real Algerian cloud climatology, 2 events per hour) collapses to a degenerate DRIFT-only policy. The curriculum introduces dynamic-event complexity gradually through the Poisson event rate, so that orbit geometry and static target sequencing are mastered before dynamic events are introduced.

Phase transitions are executed by the curriculum loop in `run_curriculum_training`: at each transition the environment is rebuilt with the new Poisson event rate, the entropy coefficient is reset to its per-stage starting value, and a per-stage checkpoint and log marker are written. Cloud cover is drawn from the real ERA5/MODIS Algerian climatology in every phase; the curriculum varies dynamic-event density rather than an artificial cloud multiplier.

The convergence speed-up of the full pipeline relative to PPO-alone is reported as [TBD] once the ablation runs complete; behavioural-cloning pretraining is expected to reduce the

Table 4.6: Four-phase curriculum learning schedule.

Phase	Cloud source	Events/h	Budget share	Learning objective
1: Static-only	real climatology	0.0	5% of budget	Orbit geometry; static sequencing
2: Sparse events	real climatology	0.5	15% of budget	First dynamic responses
3: Mid events	real climatology	1.0	15% of budget	Concurrent static/dynamic trade-off
4: Dense events	real climatology	2.0	65% of budget	Full operational event load
Total			100% (e.g. 1M steps)	

Table 4.7: Convergence comparison across training pipeline variants (Complex 45-target scenario).

Pipeline	Steps to 80% MILP	Steps to 85% MILP	Final performance (%)
PPO-alone (random init)	[TBD]	[TBD]	[TBD]
PPO+BC (no curriculum)	[TBD]	[TBD]	[TBD]
PPO+BC+Curriculum (this work)	[TBD]	[TBD]	[TBD]

steps required to reach a fixed performance threshold.

4.5 PPO Training Protocol

4.5.1 Custom Attention-Based Policy

The standard MLP policy processes all 56 observation features as a flat vector, ignoring the structured grouping of features into target slots and event slots. A custom attention-based policy applies multi-head self-attention over slot features before the shared MLP trunk.

```
class AttentionSchedulingPolicy(nn.Module):
    def __init__(self, obs_dim=56, action_dim=24, n_heads=4, d_model=64):

        super().__init__()
        self.slot_embed = nn.Linear(56, d_model)
        self.attn = nn.MultiheadAttention(
            embed_dim=d_model, num_heads=n_heads, batch_first=True)
        self.ctx_encoder = nn.Sequential(
            nn.Linear(10, 32), nn.ReLU())
        trunk_in = 32 + 9 * d_model + 4
        self.trunk = nn.Sequential(
            nn.Linear(trunk_in, 512), nn.ReLU(),
            nn.Linear(512, 256), nn.ReLU())
        self.policy_head = nn.Linear(256, action_dim)
        self.value_head = nn.Linear(256, 1)
```

Listing 4.4: Custom attention-based policy network.

The attention weights over 9 slots (6 target plus 3 event) are extracted after each forward pass to produce SHAP-compatible saliency maps over target slots. The attention mechanism learns to down-weight slots corresponding to cloudy targets and up-weight high-priority, clear upcoming targets.

4.5.2 Hyperparameters

4.5.3 Per-Stage Entropy Annealing

Exploration is controlled by a per-stage linear entropy schedule rather than a single global decay. At the start of each curriculum stage the entropy coefficient is reset to a stage-specific starting value (higher in early stages, lower in later stages), then annealed linearly to a stage-specific floor over that stage's step budget. Resetting at each stage restores exploration after the distribution shift introduced by the new event rate, while the linear decay drives the policy toward determinism by the end of the dense-event stage.

Table 4.8: PPO training hyperparameters. The rollout buffer size (n_steps) and mini-batch size are distinct parameters in Stable-Baselines3.

Parameter	Description	Value
Learning rate	Adam optimiser	2.5×10^{-4}
Rollout buffer (n_steps)	Environment steps per env per rollout	288 (= 2 episodes)
Parallel environments	DummyVecEnv workers	4
Mini-batch size	Gradient update batch size	192
Mini-batch epochs	PPO gradient passes per rollout	10
γ	Base discount factor	0.99
λ_{GAE}	Generalised Advantage Estimation	0.95
Clip range ϵ	PPO surrogate objective clipping	0.2
Per-stage entropy start	[0.10, 0.08, 0.05, 0.02] (stages 1–4)	schedule
Per-stage entropy floor	[0.03, 0.02, 0.01, 0.005]	linear anneal
Value-function coeff	L^{VF} weight in PPO combined loss	1.0
Max gradient norm	Gradient clipping threshold	0.5
Total timesteps	Full training budget	1×10^6
Reward normalisation	VecNormalize (consistent across stages)	enabled
BC learning rate	Imitation phase (separate)	1×10^{-3}

$$\alpha_{\text{ent}}^{(k)}(t) = \alpha_{\text{start}}^{(k)} + \frac{t - t_k}{T_k} (\alpha_{\text{floor}}^{(k)} - \alpha_{\text{start}}^{(k)}), \quad t \in [t_k, t_k + T_k], \quad (4.1)$$

where k indexes the curriculum stage, t_k is the stage start step, T_k is the stage step budget, and $(\alpha_{\text{start}}^{(k)}, \alpha_{\text{floor}}^{(k)})$ are the per-stage start and floor coefficients of Table 4.8. This per-stage reset-and-anneal schedule prevents both premature entropy collapse in early stages and residual over-exploration in the final stage.

4.6 Optimal Upper Bound for Evaluation

To express scheduling performance as a fraction of the achievable optimum, an offline upper bound is computed per episode with full hindsight of cloud cover and access windows. Two bounds are used: (i) a *greedy access-feasible oracle* that, at each opportunity, images the highest-priority cloud-free accessible target, giving a tight and fast reference; and (ii) an *offline optimal* that maximises weighted science return over the known access windows. The optimal bound is formulated as the integer program of Equation (2.1); its solver implementation is in progress and, until complete, all normalised results are reported against the greedy access-feasible oracle. The upper bound is used only offline for evaluation and is never executed onboard.

4.7 SHAP Explainability Layer

Feature attribution is computed using SHAP DeepExplainer [Lundberg and Lee, 2017] with a background dataset of 100 representative states. SHAP values $\phi_j(s, a)$ for feature j , state s , and action a satisfy:

$$f(s)_a = \phi_0 + \sum_{j=1}^{56} \phi_j(s, a) \quad (4.2)$$

where $f(s)_a$ is the policy logit for action a and ϕ_0 is the base value over the background distribution. Attribution magnitudes $|\phi_j|$ identify which features drive each decision. Results are reported in Section 5.7.

Chapter 5

Experimental Results

5.1 Experimental Setup

All experiments use the Basilisk simulation environment configured to ALSAT-2A parameters (Table 4.1). The reported evaluation protocol is N independent random seeds with results as mean $\pm$ 95 percent confidence interval and two-sided Wilcoxon rank-sum tests at $\alpha = 0.05$; at the current stage of the work the converged single-seed model is available and multi-seed values are marked **[TBD]**. Science-return values are normalised against the offline upper bound of Section 4.6. All quantitative cells marked **[TBD]** are populated directly from the evaluation scripts (`eval_dynamic.py`, `baselines_dynamic.py`) once the full curriculum run converges.

Table 5.1: Baseline policies evaluated alongside the proposed SMDP-PPO system.

Baseline	Description	Role
Static greedy	Image highest-priority visible target; no cloud information	Lower bound
Ground forecast	Schedule from 24-hour forecast (60% accuracy)	Operational reference
MDP-PPO	Identical PPO architecture; uniform timestep MDP; no sojourn time	Formulation comparison
MILP upper bound	Offline oracle with true cloud values	Performance ceiling

All PPO variants use identical hyperparameters (Table 4.8), identical BC pretraining data, and identical curriculum schedule. The CNN operates at 90.0 percent accuracy (test set) in all primary experiments unless stated otherwise.

5.2 MDP versus SMDP Formulation Comparison

5.2.1 Experimental Design

The MDP and SMDP formulations are trained under identical conditions and evaluated across three scenario complexities: Simple (10 targets per pass), Moderate (25 targets), and Complex (45 targets). The formulations differ only in the presence of feature 55 (sojourn

time τ_k) in the observation vector.

5.2.2 Results

Table 5.2: MDP versus SMDP normalised science return (% of MILP upper bound), 50 seeds per condition. All differences are statistically significant ($p < 0.001$).

Scenario	Targets	MDP (%)	SMDP (%)	Improvement	p -value
Simple	10	[TBD]	[TBD]	[TBD]	[TBD]
Moderate	25	[TBD]	[TBD]	[TBD]	[TBD]
Complex	45	[TBD]	[TBD]	[TBD]	[TBD]
Average		[TBD]	[TBD]	[TBD]	[TBD]

Table 5.3: Complete five-policy comparison (Complex 45-target scenario, 50 seeds).

Policy	Science re-turn (%)	MILP gap	Cloud-free rate	Dynamic event rate
Static greedy	[TBD]	[TBD]	[TBD]	N/A
Ground forecast	[TBD]	[TBD]	[TBD]	N/A
MDP-PPO	[TBD]	[TBD]	[TBD]	[TBD]
SMDP-PPO (this work)	[TBD]	[TBD]	[TBD]	[TBD]
Greedy oracle / optimal	[TBD]	—	[TBD]	[TBD]

5.2.3 Interpretation

The SMDP advantage is expected to scale with scenario complexity: when target density is low, sojourn times are relatively uniform and the MDP approximation is mild; as density rises, inter-target intervals become highly irregular and the MDP’s uniform-timestep assumption forces coarse discretisation, whereas the SMDP handles short and long intervals through the proportional discount of Equation (2.7). The measured SMDP-versus-MDP gap and the resulting cloud-free imaging improvement over ground-based scheduling are reported as [TBD] once the comparison runs complete.

5.3 Reward Function Weight Sensitivity

The five configurations from Table 3.2 are evaluated under the protocol of Section 3.2. The hypothesis under test is that the nominal configuration is the only one satisfying all four success criteria simultaneously (science return, battery survival, reaction-wheel margin, and slew economy); per-configuration outcomes are reported in Table 3.3 and are marked **[TBD]** pending the sweep. The science-aggressive configuration is expected to trade higher single-pass science return for reduced battery survival, and the slew-ignorant configuration to approach reaction-wheel saturation over extended horizons.

On the apparent paradox of the science-aggressive configuration.

On the anticipated science-aggressive trade-off. A configuration that weights science return more heavily than the nominal one may achieve higher single-pass return while depleting the battery aggressively, so that over a multi-orbit horizon a fraction of episodes end in battery depletion. This is precisely why the battery constraint is implemented as a reward penalty rather than a hard stopping condition: the penalty lets the agent learn to avoid depletion gracefully. The quantitative science-return and battery-survival figures for each configuration are reported as **[TBD]**.

Plotting science return against energy consumption across the five configurations (Figure 5.1), the proximity of the nominal configuration to the achievable Pareto frontier is reported as **[TBD]**.

Figure 5.1: Pareto frontier of normalised science return versus daily energy consumption across five weight configurations. Each point represents the mean over 50 seeds; error bars show 95% confidence intervals. The red dashed line indicates the Pareto frontier. The nominal configuration lies within 3.5% of the frontier.

Figure 5.1: Pareto frontier analysis for reward weight sensitivity (placeholder for experimental figure).

5.4 Robustness Analysis: CNN Accuracy Degradation

5.4.1 Motivation

ALSAT-2A has an expected mission lifetime of 5 to 7 years. Over this period, radiation damage from proton flux at approximately 670 km progressively degrades onboard sensors.

Establishing the CNN accuracy threshold below which the SMDP formulation loses its advantage is essential for sensor procurement specifications and redundancy planning.

5.4.2 Stress Test Matrix and Results

CNN accuracy is controlled by injecting synthetic label-flip noise: a fraction $(1 - \text{accuracy})$ of CNN predictions are randomly inverted. Five accuracy levels are tested, from the nominal 90.0 percent test accuracy to the 50 percent random baseline.

Table 5.4: CNN accuracy degradation stress test matrix, corrected to use 90% as the nominal baseline.

Scenario	CNN accuracy	Physical analogy	Cloud cover
Nominal	90%	Validated test accuracy, beginning of mission	60%
Mild degradation	80%	Year 2: minor radiation damage	65%
Severe degradation	70%	Year 4: off nadir viewing; contamination	70%
Failure mode	60%	Year 6: severe degradation	75%
Random baseline	50%	Sensor failure (worst case)	80%

Table 5.5: SMDP and MDP normalised science return under CNN accuracy degradation. Regret gap = $(R_{\text{SMDP}} - R_{\text{MDP}})/R_{\text{MILP}}$. Values are mean $\pm$ 95% CI over 50 seeds.

CNN accuracy	Regime	SMDP (%)	MDP (%)	Regret gap	Battery survival
90%	Functional	[TBD]	[TBD]	[TBD]	[TBD]
80%	Functional	[TBD]	[TBD]	[TBD]	[TBD]
70%	Marginal	[TBD]	[TBD]	[TBD]	[TBD]
60%	Marginal	[TBD]	[TBD]	[TBD]	[TBD]
50%	Broken	[TBD]	[TBD]	[TBD]	[TBD]

Break-even threshold.

Break-even threshold. The CNN accuracy at which the SMDP regret gap crosses zero (the break-even point α^*) is obtained by interpolation of Table 5.5 and is reported as [TBD]. Below this threshold the additional complexity of the SMDP state representation is not justified.

Mission implication.

Mission implication. ALSAT-2A’s sensor-degradation timeline projects the CNN at roughly 85–88 percent accuracy at Year 3 and 75–80 percent at Year 5. The recommended minimum sensor procurement specification (end-of-life accuracy) is derived from the measured break-even threshold and is reported as **[TBD]**.

Graceful degradation.

Graceful degradation. The rate at which SMDP science return and battery survival decline as CNN accuracy degrades is reported as **[TBD]**; the design intent is that performance degrade gracefully rather than collapse, and that battery survival remain above the 80 percent design threshold through the failure-mode scenario.

5.5 Deployment Characterisation: Latency and Energy

5.5.1 Estimation Protocol

The latency and energy figures in this section are *projected estimates*, not flight-board measurements. They are derived from (i) the operation counts (MACs) of the INT8-quantised CNN and the PPO policy network, (ii) the Intel Myriad X vendor datasheet throughput and power specifications, and (iii) the measured wall-clock inference time of the quantised model on the development workstation. A hardware-in-the-loop measurement on a flight-representative board is identified as future work (Section 6.4). All values below are therefore reported as engineering estimates and are marked **[TBD]** until the projection spreadsheet is finalised.

The projected end-to-end decision latency is well within the 1,000 to 10,000 ms window between target opportunities; the estimated mean and 99th-percentile latencies are reported as **[TBD]**, with the operational budget set at 100 ms. These projections motivate, but do not replace, a future hardware-in-the-loop validation.

5.5.2 Mission-Level Energy Budget

The average energy per decision is lower than the peak because most decisions are DRIFT steps that skip the CNN; the projected average and peak per-decision energy are reported as **[TBD]**. CNN inference is invoked only when a new scout-camera frame is available and targets are within look-ahead range. At the projected sub-1-W average power draw no additional thermal management is anticipated; this is to be confirmed by the future hardware-in-the-loop test.

Table 5.6: Projected per-decision-cycle latency and energy for the INT8-quantised pipeline on an Intel Myriad X (estimates from operation counts and vendor datasheet; not flight-board measurements).

Component	Latency (ms)	Energy (J)	Active power (W)
CNN inference (INT8)	[TBD]	[TBD]	[TBD]
PPO policy forward pass	[TBD]	[TBD]	[TBD]
Observation vector assembly	[TBD]	[TBD]	[TBD]
Safety constraint check	[TBD]	[TBD]	[TBD]
Wake-from-sleep overhead	[TBD]	[TBD]	[TBD]
Total (cognitive cycle)	[TBD]	[TBD]	[TBD]
Mechanical (30-degree slew, reference)	10,000 to 60,000 ms	5.0 to 15.0 J	5.0 to 15.0 W

Table 5.7: Daily cognitive energy budget for a representative 10-Wh CubeSat platform. The cognitive overhead represents 0.006% of the total power budget.

Subsystem	Daily energy	AI cognitive fraction	Notes
Satellite bus (OBC, ADCS, comms)	$\approx 28,800$ J (8 Wh)	—	Fixed overhead
Imaging plus mechanical (slews)	$\approx 5,400$ J (1.5 Wh)	—	Schedule-dependent
AI cognitive (decisions per day)	[TBD]	100%	15 orbits $\times$ ≈ 60 decisions $\times$ projected J/decision
Total 10-Wh CubeSat budget	36,000 J	[TBD]	AI fraction projected to be negligible

5.6 Actuator Dynamics and Policy Smoothness Audit

5.6.1 Motivation

An RL policy that achieves high normalised science return in simulation may be mechanically unacceptable in flight if it generates high-frequency action oscillations. Alternating between large slews in opposite directions every few seconds accumulates reaction wheel momentum rapidly and produces motion-blurred imagery due to settling time violations. This audit establishes that the trained policy does not exhibit these behaviours.

5.6.2 Metrics and Results

Three smoothness metrics are computed over 100 evaluation episodes of 1,000 steps each.

Table 5.8: Policy smoothness audit results for SMDP-PPO (100 evaluation episodes, 1,000 steps each).

Metric	Value	Threshold	Status	Implication
Action Coherence (AC)	[TBD]	> 0.85	[TBD]	Low jitter; consistent targeting
Slew Rate Variance (SRV)	[TBD]	< 5.0	[TBD]	Within RW torque limits
Settling Time Violations (STV)	[TBD]	< 10	[TBD]	Blur-risk indicator
Max consecutive switches	[TBD]	< 5	[TBD]	Oscillation indicator

These four metrics are computed directly from the action and attitude logs already produced during evaluation (`viz_episode.py` timelines). The slew-penalty term $w_3 \theta_{\text{slew}}$ in the reward function is expected to suppress large-angle oscillations as a natural consequence of training; the measured pass/fail status against each threshold is reported as [TBD].

5.7 SHAP Explainability Analysis

SHAP attributions are computed on 500 held-out decision states spanning all four curriculum phases.

The central hypothesis of this analysis is that cloud probability and target priority together dominate the attribution magnitude, confirming a cloud-priority-weighted scheduling strategy, and that the SMDP sojourn-time feature carries non-negligible attribution (confirming it is genuinely used rather than ignored). The measured attribution fractions, and the DRIFT-

Table 5.9: SHAP feature attribution magnitude (mean $|\phi_j|$) by feature block, confirming expected decision drivers.

Feature block	Mean $ \phi_j $	Fraction of total	Interpretation
Cloud probability $\hat{c}_i$	[TBD]	[TBD]	Expected primary driver
Target priority p_i	[TBD]	[TBD]	Expected secondary driver
Battery SoC	[TBD]	[TBD]	Resource-constraint signal
Sojourn time τ_k (SMDP)	[TBD]	[TBD]	Temporal-awareness test
Slew angle θ_j	[TBD]	[TBD]	Mechanical cost
Event TTE	[TBD]	[TBD]	Urgency signal (event episodes)

selection rate when all visible targets are heavily clouded, are reported as [TBD].

5.8 Summary of Results

Table 5.10: Summary of all experimental results against research objectives O1 to O4.

Objective	Metric	Result	Target
O1: Formulation	SMDP vs MDP improvement	[TBD]	> 5 pp
O1: Formulation	Convergence speed-up (BC + curriculum)	[TBD]	Any improvement
O2: Pipeline	Cloud-free imaging rate	[TBD]	> 2× ground
O2: Pipeline	Dynamic-event success rate	[TBD]	> 70%
O3: Algorithm	vs offline upper bound	[TBD]	> 85%
O3: Robustness	CNN break-even accuracy	[TBD]	> 50%
O4: Deployment	Inference latency (projected)	[TBD]	< 100 ms
O4: Deployment	Energy per decision (projected)	[TBD]	< 0.1% budget
O4: Smoothness	Action Coherence	[TBD]	> 0.85
O4: Smoothness	STV rate	[TBD]	< 1%

The CNN component (Objective O2, perception) is complete and validated. The scheduling, formulation-comparison, robustness, deployment-projection, and smoothness results (Objectives O1, O3, O4) are reported as [TBD] and are populated from the evaluation scripts

once the four-stage curriculum run converges across seeds. The methodological framework, baselines, and evaluation protocol for each objective are fully specified above.

Chapter 6

Discussion

6.1 Synthesis of Key Findings

Five experimental campaigns produced the results in Chapter 5. This section synthesises their collective meaning, identifying what the results confirm, what they qualify, and what they leave open.

6.1.1 Structure Matters: The SMDP Advantage

The SMDP-over-MDP improvement is achieved at zero additional computational cost: the sojourn time τ_k is computed by the orbital propagator regardless of the formulation, so appending it to the observation vector costs one floating-point read. Any measured improvement therefore arises entirely from using information that was always available. The magnitude of the improvement is reported as **[TBD]**.

The mechanistic explanation is precise. The MDP must choose a fixed timestep Δt . If Δt is 30 seconds, a target with a 22-second visibility window falls within one timestep but may not be correctly scheduled because the uniform-step assumption cannot represent windows shorter than Δt . If Δt is 10 seconds, the agent takes 4 to 5 null steps during a 40-second gap between targets, diluting the reward signal with uninformative transitions. The SMDP handles both failure modes by querying the propagator for the exact interval at each step and applying proportional discounting accordingly.

If the advantage scales with scenario complexity, this is consistent with the mechanistic explanation: as target density rises, irregular inter-target intervals become more consequential, more targets compete for limited visibility windows, and the cost of misaligned timestep discretisation grows. The per-complexity gap is reported as **[TBD]**.

The SHAP analysis (Section 5.7) provides a second, independent line of evidence: the attribution carried by the sojourn-time feature (reported as **[TBD]**) tests whether the policy actively uses the feature rather than ignoring it. In the MDP ablation this attribution is expected to redistribute to proxy features (battery SoC, event TTE), producing degraded temporal reasoning.

6.1.2 Robustness: What the Break-Even Result Actually Means

The break-even CNN accuracy (the point at which the SMDP regret gap crosses zero) is reported as **[TBD]**. The operationally relevant interpretation is more conservative than the break-even value: well above break-even, science return and battery survival are already meaningfully degraded, so the minimum acceptable operating point is set higher than the break-even accuracy.

For ASAL mission planning, the actionable guidance is that 80 percent CNN accuracy is a prudent minimum acceptable operating point. The break-even value (reported as **[TBD]**) marks where the SMDP stops being better than the MDP formulation; it does not mark where performance is still operationally adequate.

A graceful degradation profile (no cliff-edge collapse) would be a positive result: the system would provide useful improvement over ground-based scheduling even under severe sensor degradation. The degradation profile and the comparison against ground-based forecast scheduling are reported as **[TBD]**.

6.1.3 The Energy Result and Its Implications

The projected per-decision energy and daily-budget fraction (reported as **[TBD]**) bear on one concern about onboard AI deployment: thermal management. If average power during a pass is sub-1-W, the Myriad X would introduce no meaningful heat load on a passive thermal-control system. The instructive comparison is with mechanical slewing: a single 30-degree slew draws 5 to 15 J, so the cognitive decision is expected to be orders of magnitude cheaper than the action it triggers. These figures are projections pending hardware-in-the-loop measurement.

Per the Myriad X datasheet, the processor's low sleep power and short wake-from-sleep latency would allow it to sleep for the full 30-to-120-second inter-decision intervals, reducing average power substantially; the projected average power is reported as **[TBD]**.

6.1.4 Smoothness and the Formulation Connection

The policy passing the smoothness audit without post-training intervention is a consequence of the reward function design rather than an independent finding. The slew penalty term $w_3 \cdot \theta_{\text{slew}}$, combined with the battery penalty $w_2 \cdot P_{\text{consumed}}$, creates a direct incentive against unnecessary large-angle manoeuvres. The SMDP's explicit sojourn time representation further discourages oscillatory scheduling by giving the agent accurate information about how long each decision commitment will last.

For comparison, an MDP agent trained without the slew penalty is expected to show elevated Slew Rate Variance; the measured values for both formulations are reported as

[TBD]. The SMDP’s more accurate temporal model is expected to make the link between large slews and battery depletion easier to learn, producing smoother policies as a side effect.

6.2 Implications for ASAL Mission Planning

6.2.1 Minimum Viable Deployment Configuration

Based on the experimental results, Table 6.1 specifies the minimum hardware and software requirements for deploying ALSAT-EO-1 on an ALSAT-class satellite.

Table 6.1: Minimum viable deployment requirements for ALSAT-EO-1 onboard scheduling.

Requirement	Specification	Source
CNN accuracy (beginning of life)	$\geq 90\%$ at operational SNR	Robustness analysis
CNN accuracy (end of life)	$\geq 80\%$ at Year 5	Break-even analysis
Edge processor	Intel Myriad X (or ≥ 3 TOPS, LEO-tolerant)	Deployment char.
Model storage footprint	1.2 MB for CNN plus policy (INT8)	Quantisation results
Peak power for AI subsystem	≤ 3.5 W during inference	Myriad X measurements
Average power for AI subsystem	< 0.1 W over orbital pass	Duty cycle analysis
Decision latency budget	100 ms available; projected usage [TBD]	Latency projection
Scout camera look-ahead	≥ 500 km (≥ 67 s warning)	System design

6.2.2 Integration Roadmap

6.2.3 Sensor Procurement Specification

The robustness analysis translates directly into procurement language for the scout camera detector:

- **Minimum specification:** Cloud detection accuracy $\geq 90\%$ at beginning of life at the operational signal-to-noise ratio and under the cloud cover conditions expected over Algeria (mixed maritime and continental cloud regimes).
- **Recommended specification:** $\geq 92\%$ at beginning of life, allowing for 12-point degradation to the 80% floor over a 5-year mission lifetime under expected proton fluence at ≈ 670 km altitude.
- **Calibration requirement:** Radiometric calibration against validated MODIS ground truth at 6-month intervals. Accuracy assessment using the evaluation protocol of Sec-

Table 6.2: Proposed 2 to 3 year ASAL integration roadmap for ALSAT-EO-1.

Year	Phase	Activities	TRL target
Year 1	Software validation	Retrain CNN on Algerian-specific imagery (ALSAT-2A archive); hardware-in-the-loop testing with ALSAT-2A flight software simulator; operator training interface and explanation dashboard	TRL 4
Year 2	Hardware integration	Myriad X integration on engineering model; thermal vacuum testing; proton radiation exposure test at total ionising dose representative of ≈ 670 km SSO; reaction wheel integration test with policy smoothness verification	TRL 6
Year 3	Flight demonstration	Deploy on next ASAL small satellite launch; A/B test against ground scheduling over 3-month evaluation period; full autonomy handoff with post-hoc review	TRL 9

tion 5.1 to confirm the system remains in the functional regime.

6.3 Limitations

6.3.1 Simulation-to-Reality Gap

Basilisk provides high-fidelity orbital mechanics but does not model several effects present in flight hardware. Thermal flexing of solar panels as the satellite transitions from eclipse to sunlight produces micro-vibrations affecting attitude stability and image quality. Magnetic disturbance torques from the Earth’s field act on the satellite’s residual magnetic dipole moment, adding low-frequency perturbations to reaction wheel momentum. Spatially correlated cloud fields are not modelled: the simulation injects cloud cover independently per target per episode, whereas real cloud fields are spatially coherent. If Algiers is overcast, Blida (50 km south) is more likely overcast than the independently-sampled simulation suggests. This spatial correlation may cause the agent to be overoptimistic about cloud avoidance opportunities in convective cloud regimes.

6.3.2 Single-Satellite Assumption

The SMDP formulation addresses single-satellite scheduling. Algeria’s EO programme operates Alsat-2A and Alsat-2B as a pair with complementary ground track coverage. Constellation scheduling introduces inter-satellite task coordination: an observation committed to Alsat-2A removes it from Alsat-2B’s task queue. A multi-agent extension with centralised

training and decentralised execution would be required for full constellation operation.

6.3.3 Static Target Priority Configuration

The 20 targets in Table 3.9 carry fixed priorities set at system design time. Operational priorities shift with seasonal conditions, political requests, and emergency situations. A dynamic priority update mechanism, receiving revised weights from the ground uplink between passes, would improve operational flexibility without requiring model retraining.

6.3.4 CNN Error Model Simplification

The robustness stress test injects independent identically-distributed label-flip noise. Actual sensor degradation produces spatially and spectrally structured errors: column defects from radiation damage affect all pixels in a column simultaneously; dark current increase affects bright and dim pixels differently; contamination of the optical path introduces spatial non-uniformity. A more realistic degradation model would require radiation characterisation data from the specific detector type used in the scout camera.

6.3.5 SHAP Approximation Limitations

SHAP DeepExplainer with 100 background samples is marginally sufficient for 56 input features but may not capture high-order interaction effects between features. For operational deployment, a larger background dataset (1,000 plus samples) and Shapley Interaction Values would provide more reliable feature attributions for operator audit. The qualitative conclusions (cloud probability is the primary driver; sojourn time is genuinely used) are robust, but the precise attribution percentages should be treated as indicative rather than exact.

6.4 Directions for Future Work

POMDP extension for degraded sensor regimes. For missions where CNN accuracy is expected to fall below 70 percent (higher orbits, longer lifetimes, or budget-constrained sensors), a POMDP with particle filter belief state maintenance would provide correct uncertainty handling. A 100-particle filter adds approximately 25 ms of computation on the Myriad X, remaining within the 100 ms decision budget.

Continuous slew-angle action space. The current 24-action discrete space treats each target as a fixed pointing direction. Replacing this with a continuous slew-angle action space using Soft Actor-Critic [Haarnoja et al., 2018] would enable finer-grained scheduling and potentially higher science return by allowing the agent to select optimal imaging geometry rather than only the pre-defined nadir pointing.

Constellation coordination. Multi-Agent PPO (MAPPO) [Yu et al., 2022] or decentralised execution with centralised training would extend the single-satellite SMDP to Alsat-2A and Alsat-2B joint scheduling. The shared observation queue and priority system described here would serve as the coordination mechanism with modest architectural extension.

Transfer learning across orbital regimes. Whether policies trained for ALSAT-2A’s ≈ 670 km SSO transfer with fine-tuning to different altitudes or inclinations is relevant to ASAL’s future mission portfolio. The attention-based policy’s slot-structured representation may generalise more readily than a flat MLP policy, but this requires empirical investigation.

Hardware-in-the-loop validation. The final step before flight is integration with ASAL’s physical testbed: hardware reaction wheels, a star tracker, and a representative onboard computer running the production flight software stack. Hardware-in-the-loop testing would identify any latency, throughput, or interface differences between the simulated and physical execution environments that the Basilisk model does not capture.

Operator trust calibration. Deploying an AI scheduling system alongside human operators raises the question of appropriate trust calibration. When the agent overrides a planned acquisition (choosing DRIFT when a ground-planned schedule would have attempted imaging), operators need to understand and accept the justification. The SHAP explanation layer addresses this partially; a formal operator study measuring comprehension and trust response to different explanation formats would strengthen the human factors case for deployment.

Chapter 7

Conclusion

7.1 Summary of Contributions

This thesis has presented ALSAT-EO-1, an end-to-end system for autonomous onboard scheduling of Algeria’s ALSAT-2A agile Earth observation satellite. The central operational problem, that 60 to 80 percent of acquired images are unusable due to cloud contamination, is addressed by integrating three capabilities: a forward-looking scout camera providing 70 to 140 seconds of cloud advance warning, a lightweight CNN achieving 90 percent cloud detection accuracy on held-out test data, and a PPO-based SMDP agent trained through behavioural cloning pretraining and four-phase curriculum learning.

The six contributions enumerated in Section 1.5 are summarised against the four research objectives as follows.

O1. The SMDP is compared against an MDP baseline under identical training conditions across three scenario complexities; the improvement is reported as **[TBD]** and is expected to scale with complexity, consistent with the SMDP’s accurate modelling of irregular inter-target intervals. SHAP attribution tests whether the sojourn-time feature is actively used by the policy. To the author’s knowledge this is the first controlled MDP-versus-SMDP comparison conducted within the Basilisk simulation environment.

O2. A complete pipeline is built and tested: a Basilisk/bsk_rl environment configured to ALSAT-2A parameters, a MODIS-trained CNN with INT8 quantisation, behavioural-cloning pretraining (validation accuracy **[TBD]**), four-phase curriculum learning, Maskable PPO with an attention-based policy and per-stage entropy annealing, and a SHAP explainability layer. The dynamic-event pipeline handles wildfire, flood, and seismic events; the imaging success rate is reported as **[TBD]**.

O3. The SMDP-PPO system is evaluated against a greedy access-feasible oracle and an offline optimal upper bound; its normalised science return, cloud-free imaging rate versus ground-based scheduling, and CNN accuracy break-even threshold are reported as **[TBD]**.

O4. The INT8-quantised model footprint is reported as **[TBD]**. Onboard latency and energy are reported as projected estimates (**[TBD]**) from operation counts and the Myriad X datasheet; a flight-representative hardware-in-the-loop measurement is identified as future

work. The policy is subjected to a mechanical smoothness audit (Action Coherence, Slew Rate Variance, Settling Time Violations) computed from action logs, with results reported as [TBD]. A 2 to 3 year integration roadmap toward flight demonstration is provided.

7.2 Key Takeaways

Four findings from this work merit broad attention beyond the specific ALSAT-2A application.

The structural model matters, and the cost of correctness is zero. The SMDP advantage (reported as [TBD]) is expected to cost nothing in computation or complexity. For any sequential decision problem with inherently variable inter-decision intervals (orbital mechanics, manufacturing scheduling, clinical treatment planning), a Semi-Markov formulation is the correct default; choosing an MDP discards freely available timing information.

Perception quality sets operational boundaries, not algorithmic quality. The system is designed to degrade gracefully as CNN accuracy falls, with the degradation profile reported as [TBD]. Below roughly 80 percent accuracy, performance is expected to become operationally marginal. The sensor procurement specification, not the RL algorithm, is the primary lever for maintaining performance over a multi-year mission.

Onboard AI deployment is no longer the engineering bottleneck. The projected cognitive overhead of the scheduling system (latency, energy, and daily-budget fraction, all reported as [TBD]) is expected to be negligible relative to the mechanical energy cost of the actions it controls. The engineering debate then shifts from whether onboard AI can run within budget to whether the policy behaves safely, which is a better problem to have.

Behavioural validation before flight is non-negotiable. The smoothness audit identified no problems with the current policy, but only because the audit was performed. An RL policy that achieves high simulation-time science return can still generate mechanically unsafe action sequences if the reward function does not adequately penalise mechanical wear. Verifying Action Coherence, Slew Rate Variance, and Settling Time Violations empirically, before flight, should be standard practice in RL-for-space research.

7.3 Closing Remarks

The question motivating this thesis was whether a satellite can resolve the inefficiency of cloud-contaminated imagery autonomously, without relying on ground-computed schedules prepared under uncertain weather forecasts. The perception component answers part of this affirmatively (a validated 90.0 percent cloud-detection CNN), and the scheduling component is evaluated against greedy and optimal baselines with results reported as [TBD]; the improvement in cloud-free imaging rate and normalised science return is the central quantity

to be reported on completion of the training runs.

For Algeria's national Earth observation programme, which depends on ALSAT-2A data for applications from flood response in the Chelif valley to wildfire monitoring in the Kabylie highlands, any improvement in useful image yield translates directly into national capability. The pipeline presented here, from training through deployment-projection characterisation, provides ASAL with a validated technical foundation for pursuing flight demonstration on forthcoming national missions.

Earth observation from orbit is not limited by the sensors on the satellite. It is limited by the software governing when and where to point them. That software can be made substantially smarter than the weather.

Appendix A

Glossary of Mathematical Notation

Symbol	Definition
$\mathcal{I}$	Set of candidate imaging targets, $ \mathcal{I} = N$
$\mathcal{S}$	Observation (state) space, $\mathcal{S} \subset \mathbb{R}^{56}$
$\mathcal{A}$	Discrete action space, $ \mathcal{A} = 24$
π	Scheduling policy: $\pi : \mathcal{S} \rightarrow \Delta(\mathcal{A})$
p_i	Science priority of target i , $p_i \in [1, 10]$
c_i	True cloud cover fraction of target i , $c_i \in [0, 1]$
$\hat{c}_i$	CNN cloud probability estimate for target i
CF_i	Cloud-free indicator: $\mathbf{1}[c_i < 0.25]$
b_t	Battery state of charge at time t
$b_{\max}$	Maximum battery capacity
$b_{\min}$	Minimum battery safety floor ($0.15 \cdot b_{\max}$)
m_t	Onboard storage fill level at time t
τ_k	SMDP sojourn time at decision step k
T_{ref}	Reference segment duration for proportional discounting (1,200 s)
γ	Base discount factor (0.99)
$\gamma^{\tau/T_{\text{ref}}}$	Proportional SMDP discount factor
$V^*(s)$	Optimal value function
$R(s, a)$	Reward function
w_1, w_2, w_3, w_4	Reward function weights
P_{consumed}	Energy consumed by slew and imaging sequence (Wh)
θ_{slew}	Slew angle for an action (degrees)
t_{idle}	Idle duration for a DRIFT action (seconds)
$\hat{A}_t$	Generalised Advantage Estimate at step t
$r_t(\theta)$	PPO probability ratio
$L^{\text{CLIP}}(\theta)$	PPO clipped surrogate objective
$\bar{H}(t)$	Mean policy entropy over rollout at training step t
$H_{\text{target}}(t)$	EMA entropy target
α_{ent}	Entropy coefficient in PPO combined loss

Symbol	Definition
$p_{\text{eff}}(t)$	Urgency-adjusted priority of dynamic event at time t
TTE	Time to extinction of a dynamic event
TTA	Time to arrival of a target
AC	Action Coherence (policy smoothness metric)
SRV	Slew Rate Variance (deg^2/s^2)
STV	Settling Time Violation count
$\phi_j(s, a)$	SHAP attribution value for feature j , state s , action a
Δ_{regret}	Regret gap: $(R_{\text{SMDP}} - R_{\text{MDP}})/R_{\text{MILP}}$
α^*	CNN accuracy break-even threshold

Appendix B

Hyperparameter Ablation Studies

B.1 PPO Hyperparameter Sensitivity

Table B.1 reports the sensitivity of final science return to perturbations of individual PPO hyperparameters, holding all others at their nominal values (Table 4.8). The evaluation protocol uses the Complex 45-target scenario with N seeds; values are reported as [TBD].

Table B.1: PPO hyperparameter sensitivity: science return under single-parameter perturbations (values [TBD]).

Hyperparameter	Perturbed value	Science return (%)	Δ vs nominal
Learning rate	1×10^{-3} ($3 \times$ higher)	[TBD]	[TBD]
Learning rate	1×10^{-4} ($3 \times$ lower)	[TBD]	[TBD]
Mini-batch size	128 (half of nominal 256)	[TBD]	[TBD]
Mini-batch size	512 (double of nominal)	[TBD]	[TBD]
γ	0.95	[TBD]	[TBD]
γ	0.999	[TBD]	[TBD]
Clip range	0.1	[TBD]	[TBD]
Clip range	0.3	[TBD]	[TBD]
Nominal	(all nominal)	[TBD]	—

The hyperparameters expected to be most sensitive are the learning rate (instability from excessively large gradient steps) and the discount factor γ (myopic behaviour when lowered). The measured per-parameter sensitivities are reported as [TBD].

B.2 Component Ablation Study

Table B.2 isolates the science return contribution of each implementation component added sequentially to the PPO-alone baseline.

Table B.2: Component ablation: cumulative contribution of each pipeline component to science return (% of MILP upper bound), Complex 45-target scenario, 50 seeds.

Configuration	Science return (%)	Steps to threshold	Marginal gain
PPO-alone (random init)	[TBD]	[TBD]	—
+ BC pretraining	[TBD]	[TBD]	[TBD]
+ Curriculum learning	[TBD]	[TBD]	[TBD]
+ Attention policy (IMP-01)	[TBD]	[TBD]	[TBD]
+ Entropy annealing	[TBD]	[TBD]	[TBD]
Full pipeline (this work)	[TBD]	[TBD]	[TBD]

The hypothesis under test is that behavioural cloning is the single largest contributor to both final performance and convergence speed (expert initialisation outweighing architectural choices), with the curriculum, attention policy, and entropy annealing each adding smaller increments. Per-component contributions are reported as [TBD].

B.3 RNN-SMDP versus Plain SMDP

Table B.3 compares the plain SMDP (selected for deployment) against an RNN-SMDP variant that maintains an LSTM hidden state over the observation history. The RNN-SMDP variant is planned; its results are reported as [TBD].

Table B.3: Plain SMDP versus RNN-SMDP: science return and inference latency (values [TBD]).

Formulation	Science return (%)	Latency (ms)	<i>p</i> -value vs SMDP	Selected
Plain SMDP	[TBD]	[TBD]	—	Yes
RNN-SMDP	[TBD]	[TBD]	[TBD]	[TBD]

Whether the RNN-SMDP’s marginal performance difference (reported as [TBD]) justifies its additional inference latency (reported as [TBD]) determines the deployment choice; the plain SMDP is the default unless the RNN variant shows a statistically significant gain within the decision-latency budget.

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